
Respiratory Physiology

Students : PC_1, 2025

Course objectives

At the end of this lesson, you are expected to:

- Explain mechanics of pulmonary ventilation.
- Describe pulmonary volumes and capacities.
- Discuss pulmonary circulation ,edema and pleural fluids.
- Explain physical principles of gas exchange and transport.
- Explain mechanism of regulation of respiration.

✓ Session 1:

Pulmonary Ventilation, Volumes
and Capacities

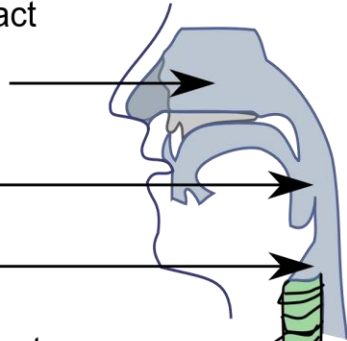
Respiratory Tract

Upper respiratory tract

Nasal cavity

Pharynx

Larynx

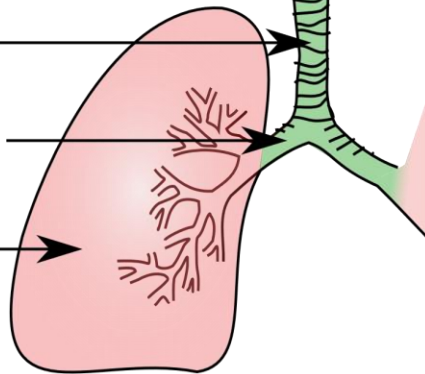


Lower respiratory tract

Trachea

Primary bronchi

Lungs



Nose

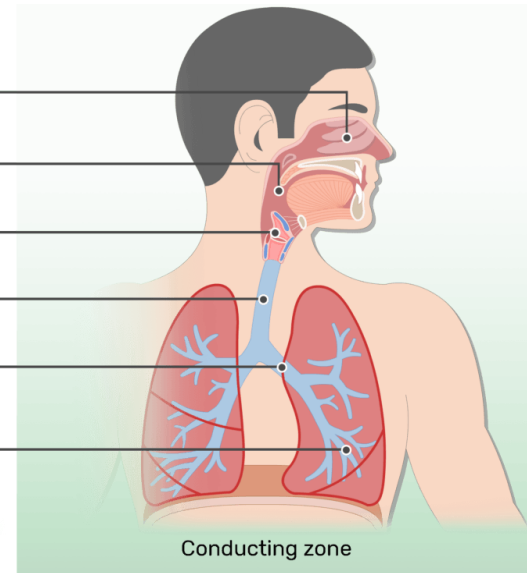
Pharynx

Larynx

Trachea

Bronchus

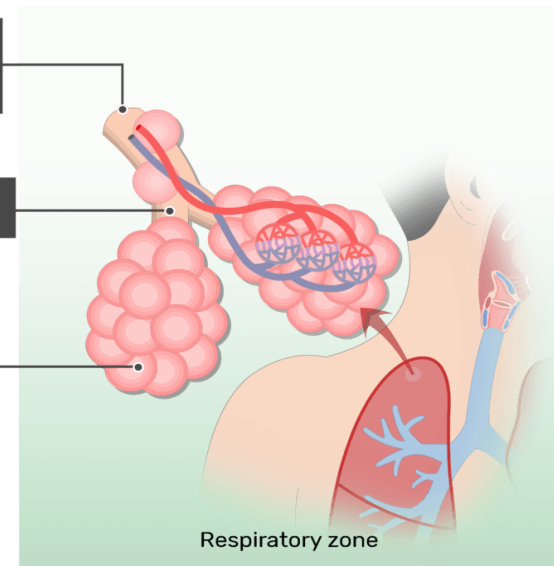
Bronchiole



Respiratory
Bronchiole

Alveolar Duct

Alveoli



What are the functions of the
respiratory system ?

Breathing

- ❖ Breathing [pulmonary ventilation] consists of two phases:

1. Inspiration/Inhalation

- air flows into the lungs.

2. Expiration/Exhalation

- gases exit the lungs.

Inspiration



Diaphragm & external intercostal muscles (inspiratory muscles) contract & the rib cage rises.



Lungs are **stretched** & intrapulmonary **volume** increases.

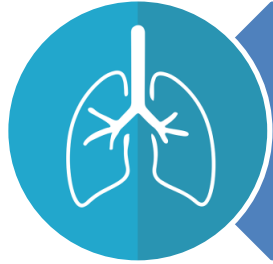


Intrapulmonary **pressure drops** below atmospheric pressure (-1 mm Hg).

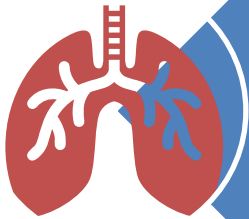


Air flows into the lungs, down its pressure gradient, until intrapulmonary pressure = atmospheric pressure.

Expiration



Thoracic cavity **volume decreases.**



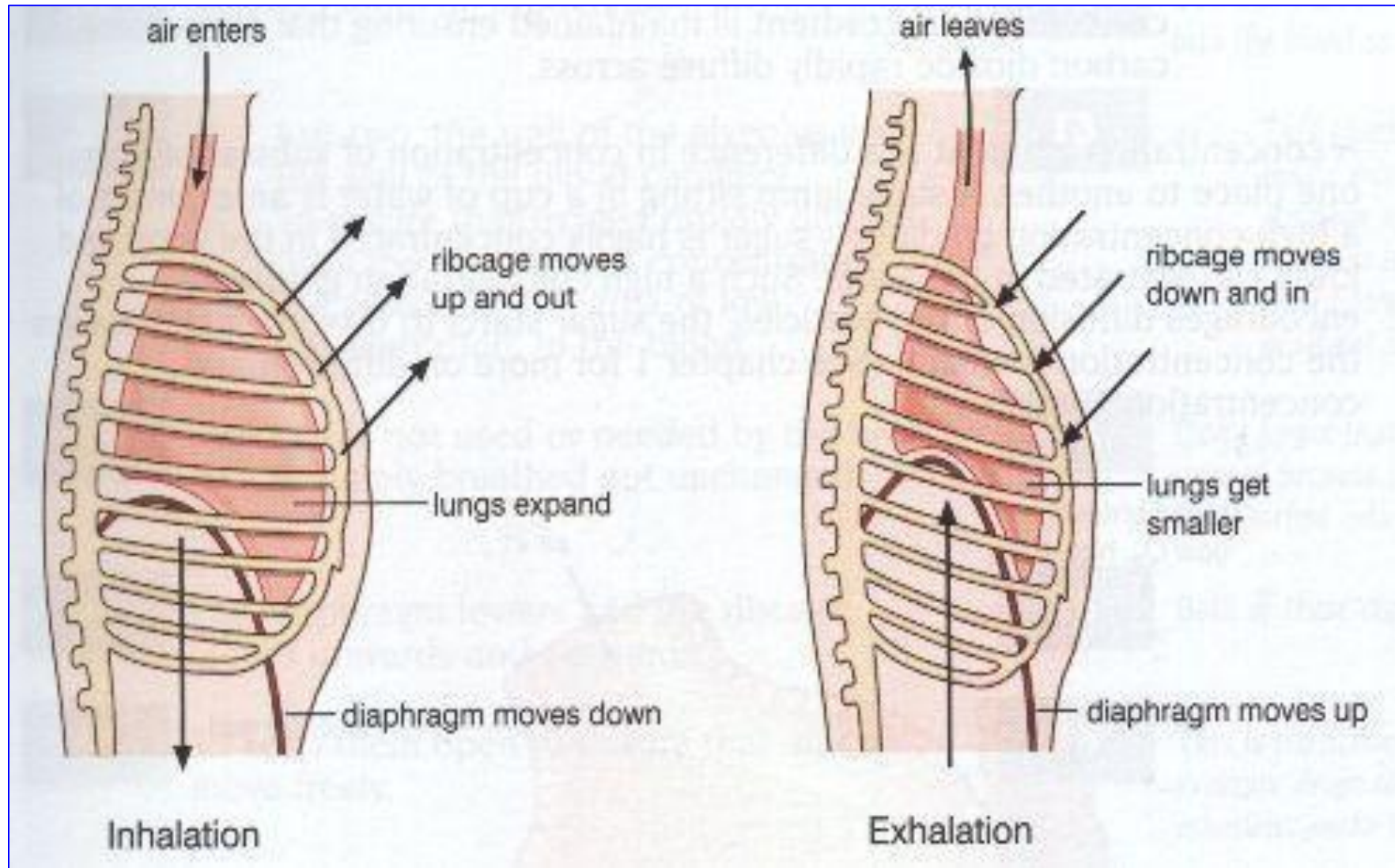
Elastic lungs recoil passively & intrapulmonary **volume decreases.**

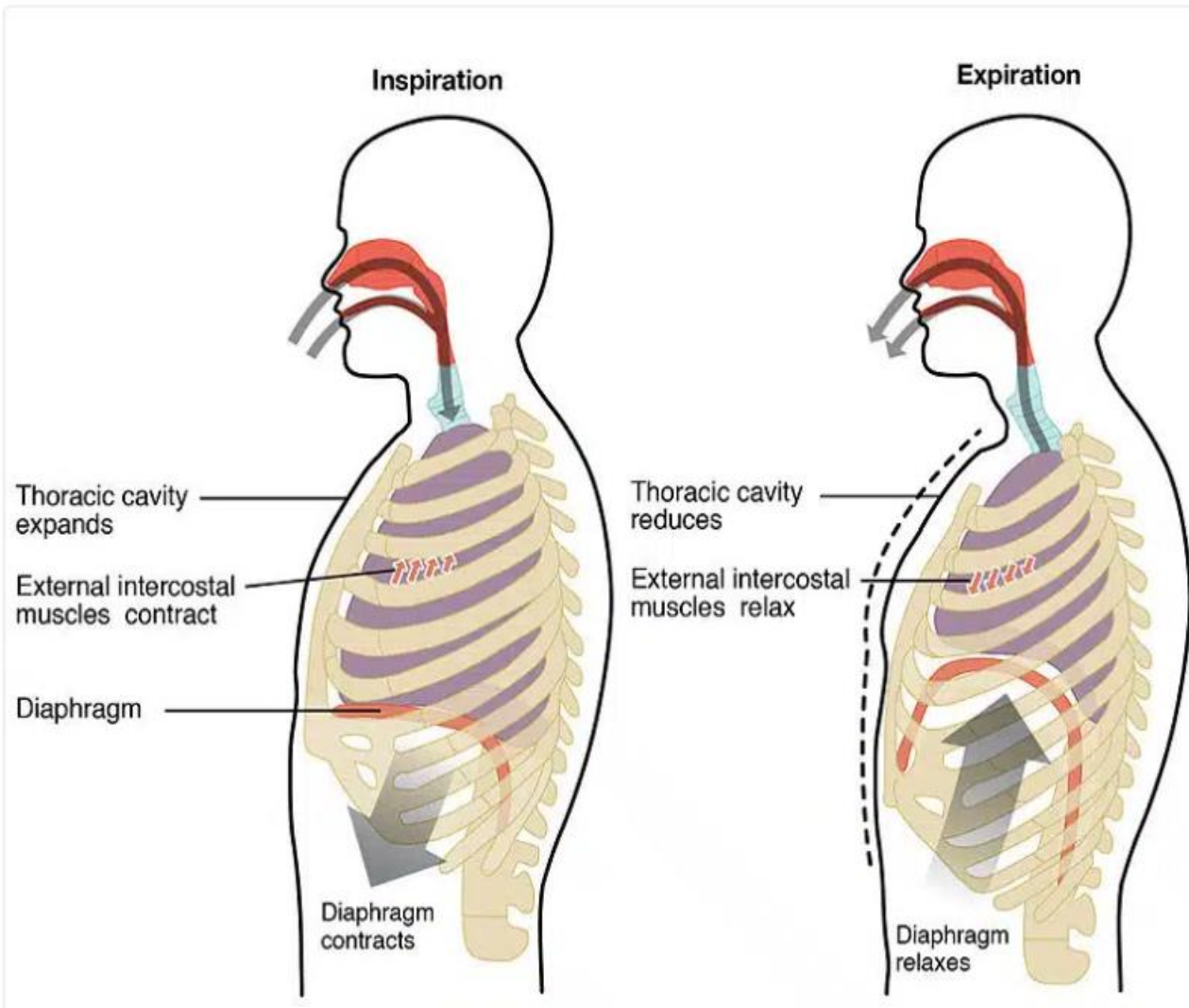


Intrapulmonary **pressure rises** above atmospheric pressure (+1 mm Hg)

✓ Gases flow out of the lungs down the pressure gradient until intrapulmonary pressure is 0.

Changes during breathing





**What about
forced breathing?**

Fig : Process of respiration at rest

MUSCLES OF INHALATION

MUSCLES OF EXHALATION

Sternocleidomastoid

Scalenes

External
intercostals

Diaphragm

Internal
intercostals

External
oblique

Internal
oblique

Transversus
abdominis

Rectus
abdominis

(a) Muscles of inhalation and their actions (left);
muscles of exhalation and their actions (right)

Sternum:

Exhalation

Inhalation

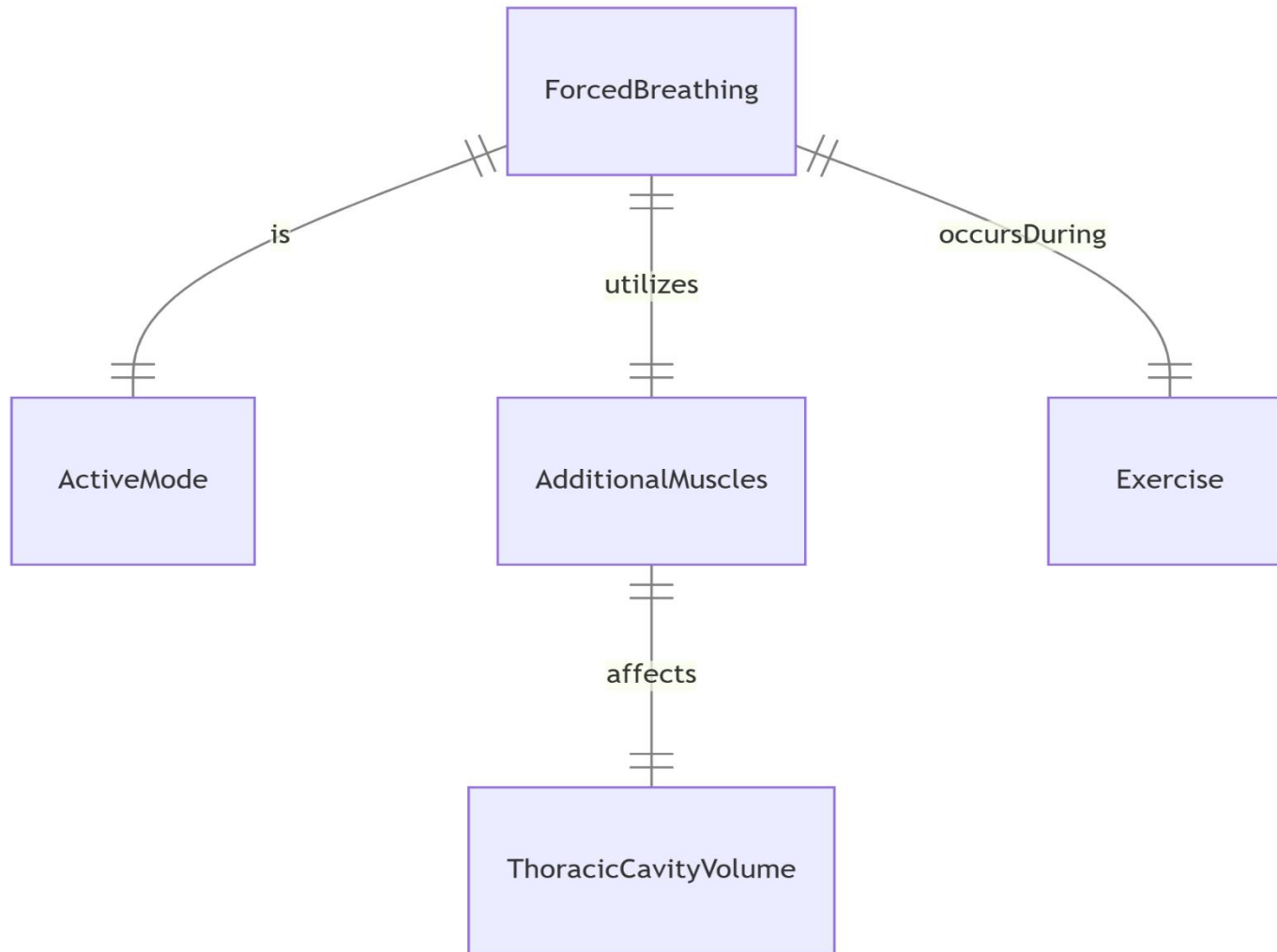
Diaphragm:

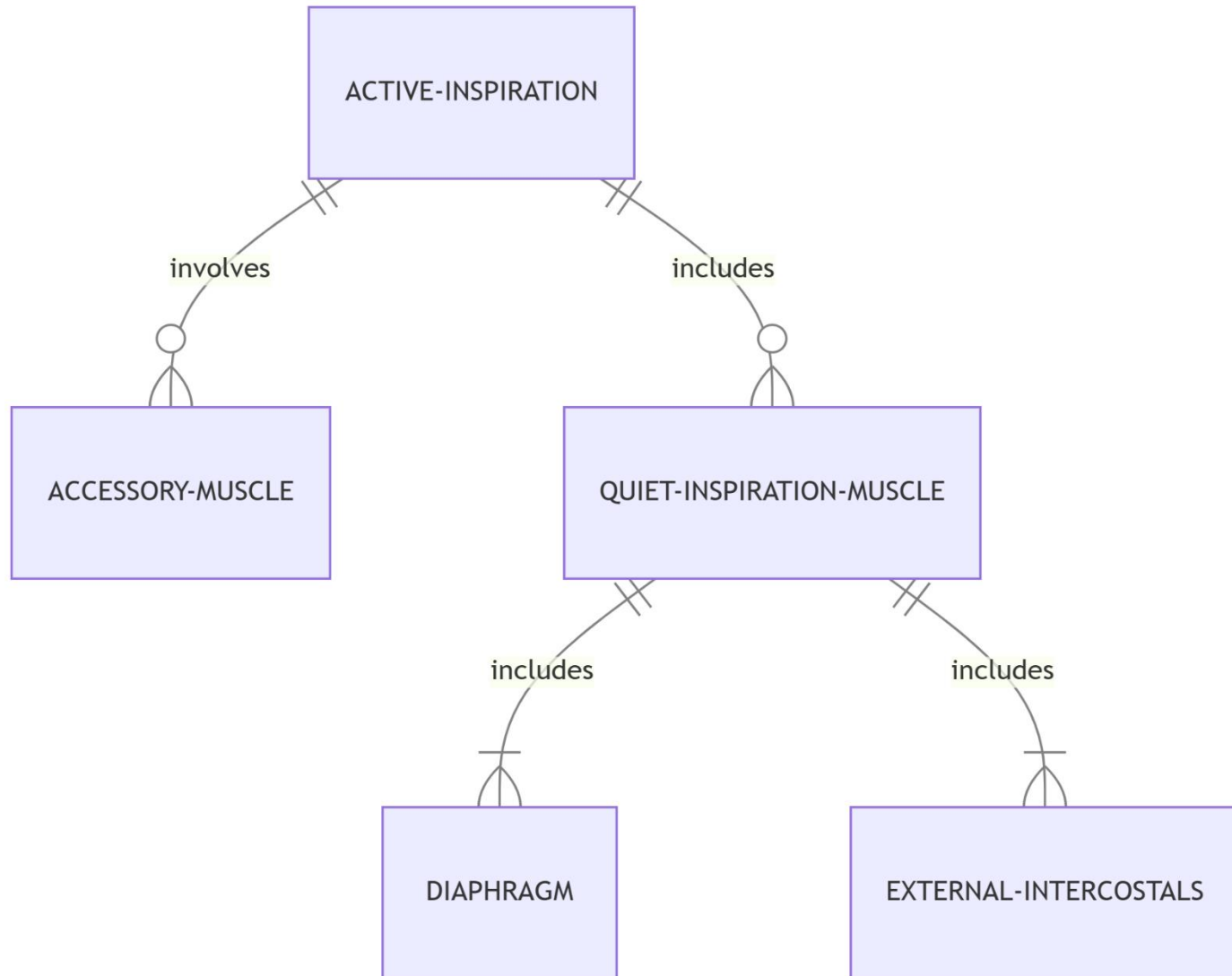
Exhalation

Inhalation

(b) Changes in size of thoracic cavity
during inhalation and exhalation

Forced breathing





Accessory inspiratory muscles

All of these muscles act to increase the volume of the thoracic cavity:



- ☐ **Scalenes** – elevates the upper ribs.
- ☐ **Sternocleidomastoid** – elevates the sternum.
- ☐ **Pectoralis major and minor** – pulls ribs outwards.
- ☐ **Serratus anterior** – elevates the ribs (when the scapulae are fixed).
- ☐ **Latissimus dorsi** – elevates the lower ribs.

Active expiratory muscles

Active Expiration

Active expiration utilises the contraction of several thoracic and abdominal muscles. These muscles act to decrease the volume of the thoracic cavity:

Transversus Abdominis and Rectus Abdominus--Push the diaphragm upward.

Internal intercostal – depresses the ribs.

External oblique--help in compressing the thoracic cavity

Internal oblique - supporting the diaphragm's upward movement

Respiratory Volumes

1. Tidal volume (TV=0.5L)
2. Inspiratory reserve volume (IRV=3L)
3. Expiratory reserve volume (ERV=1.1L)
4. Residual volume (RV=1.2L)

Lung volumes

Tidal
Volume

Volume that enters and leaves with each breath during normal quiet inspiration and expiration.

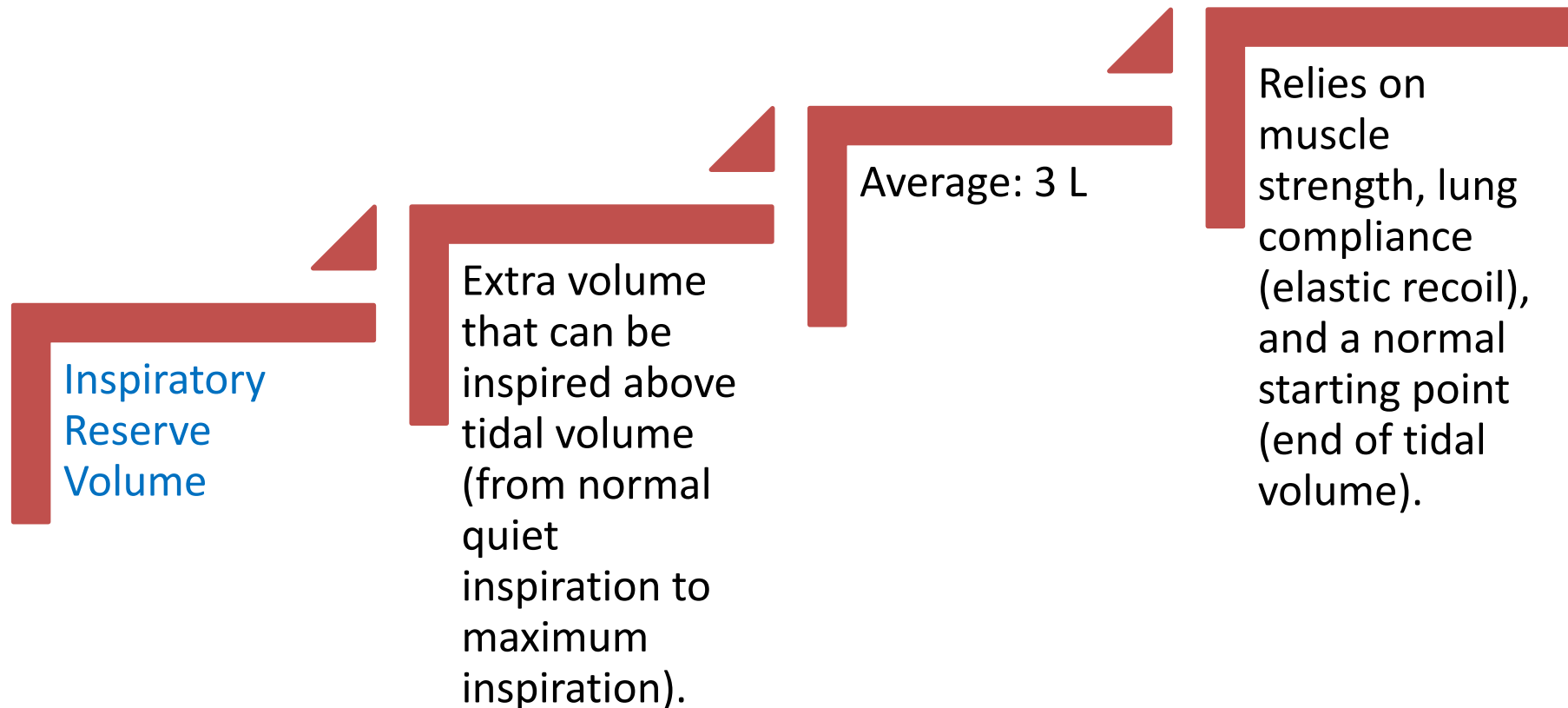
Average: 0.5
L

Changes with pattern of breathing (e.g., shallow vs. deep breaths).

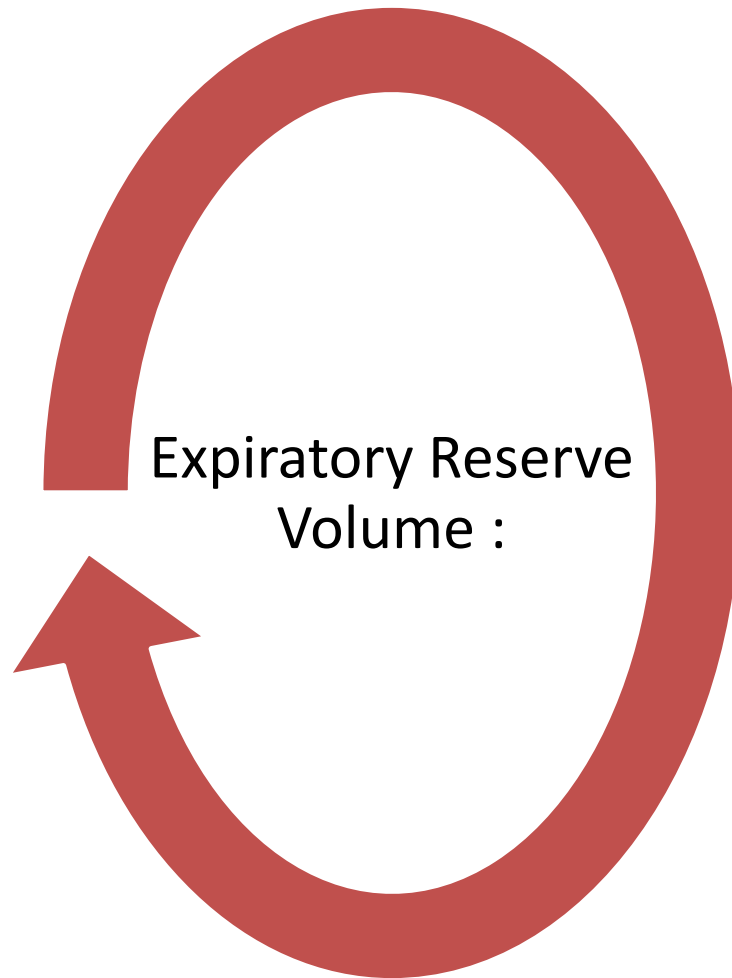
Increased during pregnancy.

Why?

Lung volumes ...

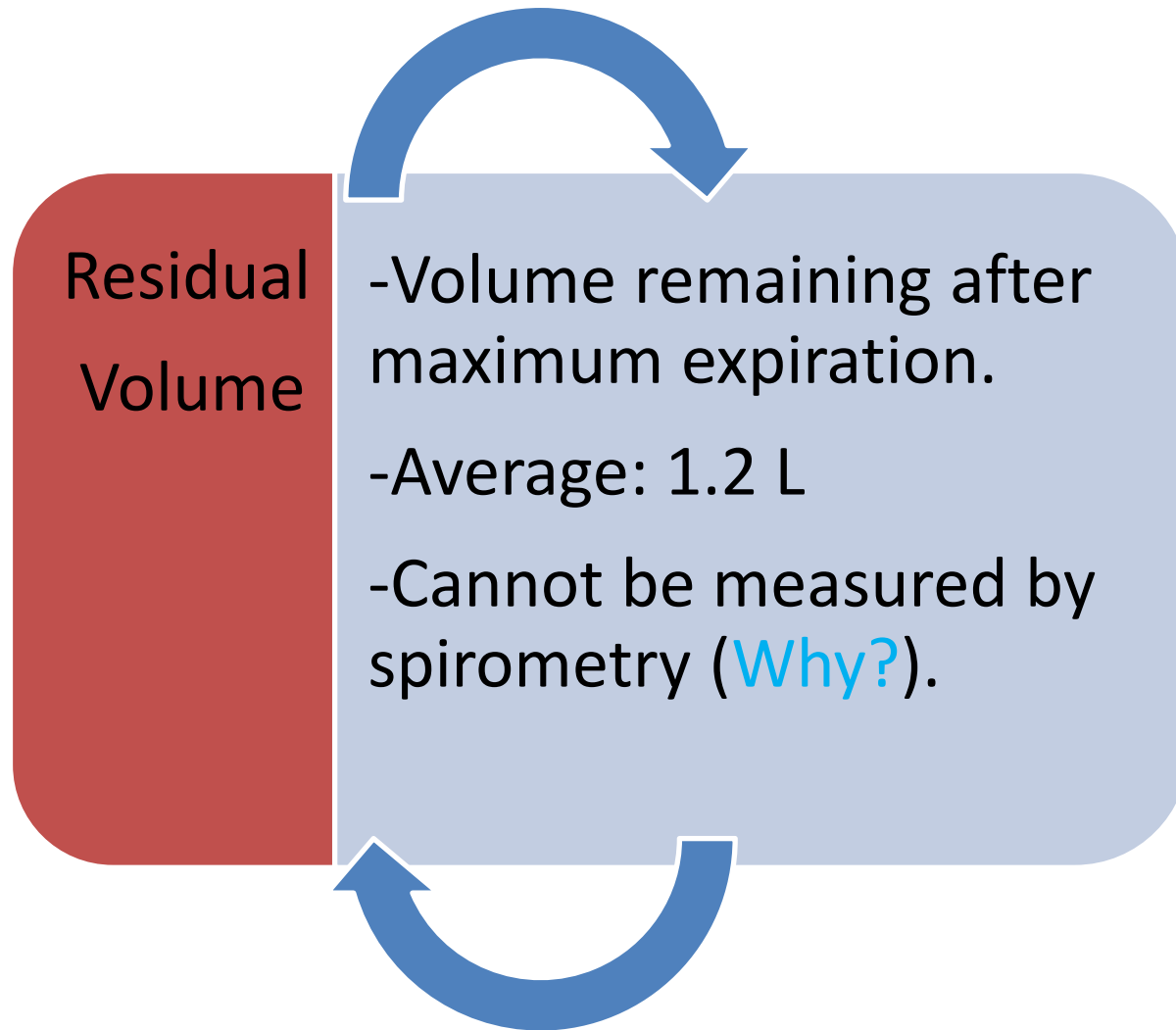


Lung volumes ...



- Extra volume that can be expired after tidal volume (from normal quiet expiration to maximum expiration).
- Average: 1.1 L
- Relies on muscle strength and low airway resistance.
- Reduced in pregnancy, obesity, severe obstruction, or proximal airway obstruction.

Lung volumes ...



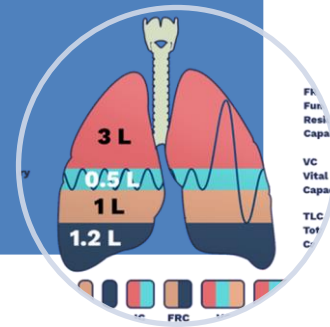
Respiratory Capacities

1. Inspiratory capacity (IC=3.5L)
2. Functional residual capacity (FRC=2.5L)
3. Vital capacity (VC=4.6L)
4. Total lung capacity (TLC=5.8L)

Lung capacities

- Volume that can be exhaled after maximum inspiration (maximum inspiration to maximum expiration).
- Inspiratory reserve volume + tidal volume + expiratory reserve volume.
- Average: 4.6 L
- Often changes in disease.
- Requires adequate compliance, muscle strength, and low airway resistance.

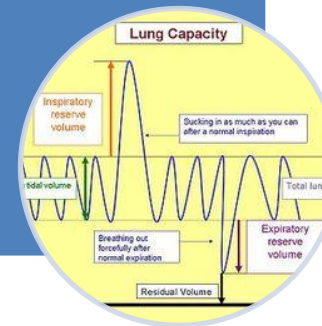
**Vital Capacity (Forced
Vital Capacity)**



Lung capacities ...

- Volume breathed in from quiet expiration to maximum inspiration.
- Tidal volume + inspiratory reserve volume.
- Average: 3.5 L

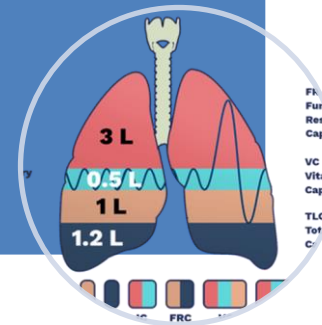
Inspiratory Capacity



Lung capacities ...

- Volume remaining after quiet expiration.
- Expiratory reserve volume + residual volume.
- Average: 2.5 L
- Affected by height, gender, posture, and changes in lung compliance.
- Height has the greatest influence.

Functional Residual Capacity

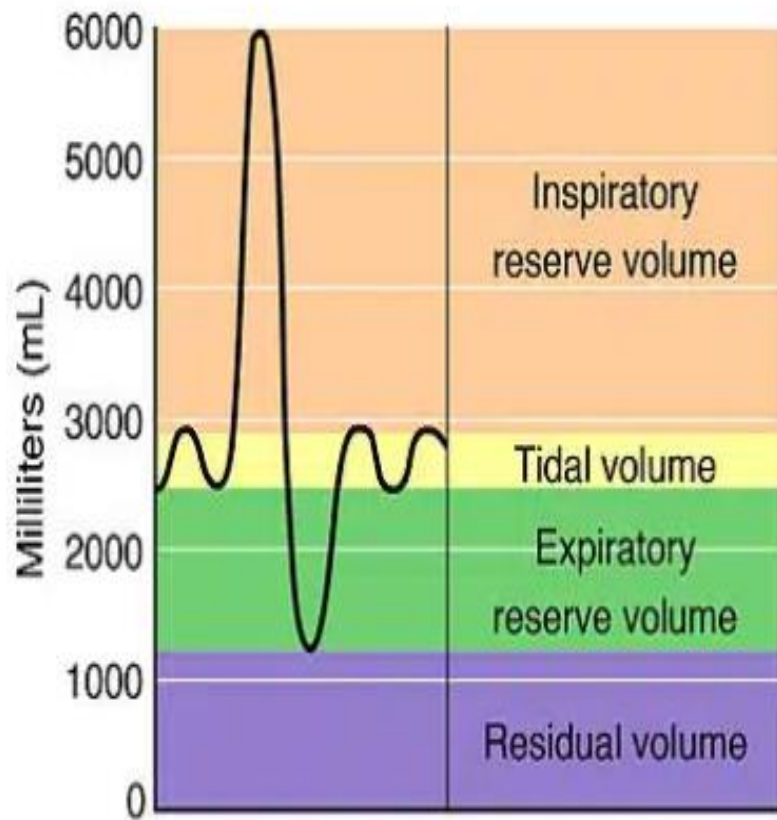


Lung capacities ...

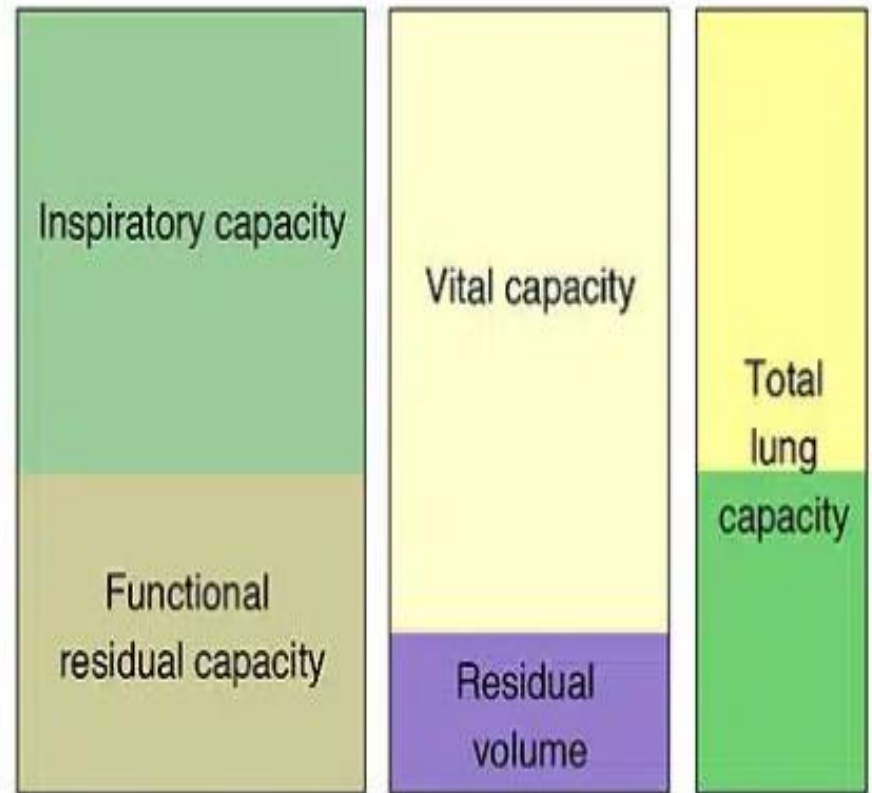
- Volume of air in lungs after maximum inspiration.
- Sum of all volumes
- $(TV + IRV + ERV + RV)$.
- **Average: 5.8 L**

Total Lung Capacity





(a)



(b)

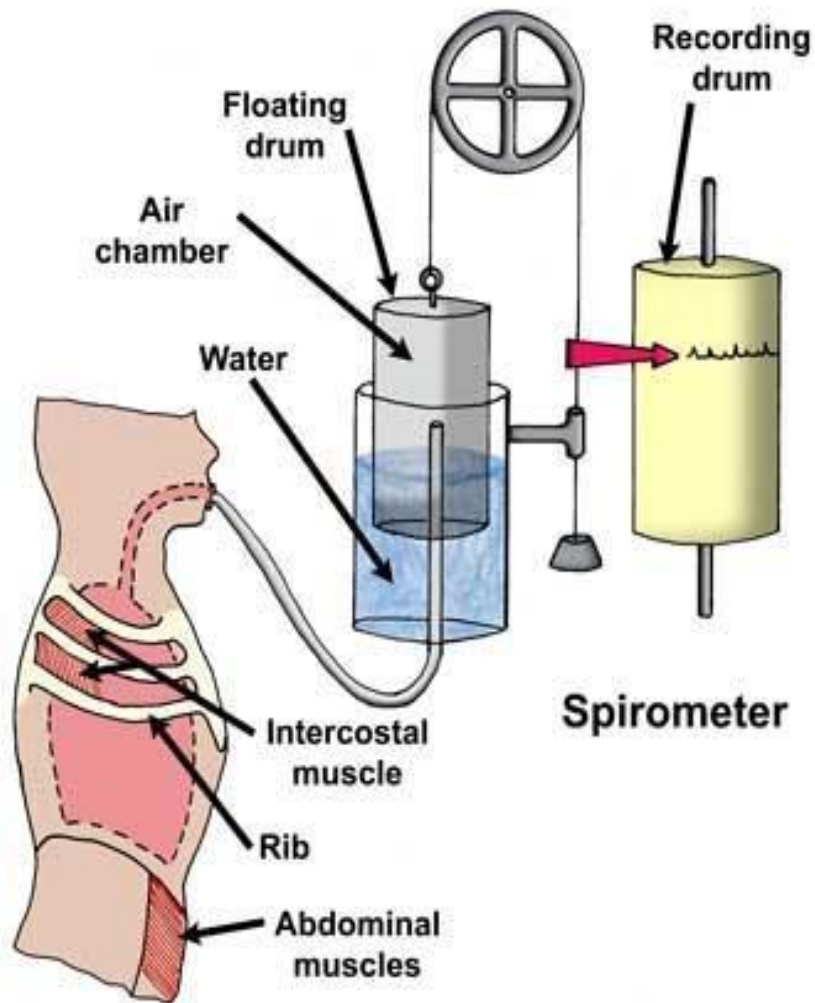
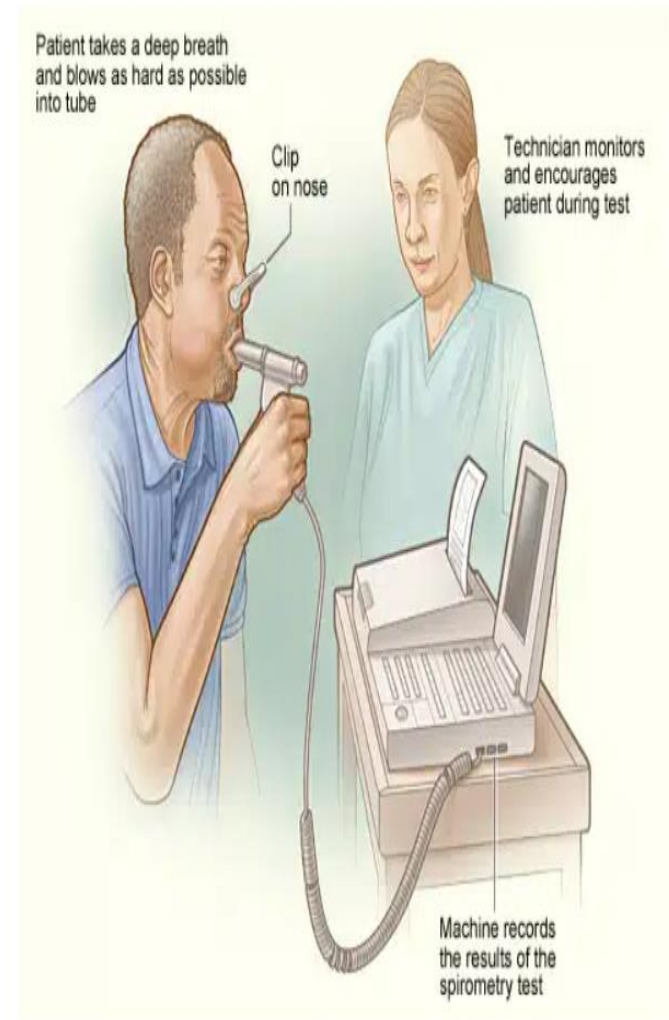


Fig :Spirometer



Visualizing lung volumes and capacities

Visualizing Lung Volumes

-Understanding lung volumes through spirometry.

What is a Vitalograph?

-A vitalograph is a device that creates plots of volume against time.
-Utilizes data collected from spirometry tests to visualize lung function.

Key Spirometry Volumes

- **Two Important Measurements:**

1)FVC (Forced Vital Capacity):

- The maximal volume of air that can be expelled in one maximal expiration from a point of maximal inspiration.

2)FEV1 (Forced Expiratory Volume in One Second):

- The maximal volume of air that can be expelled in one second from a point of maximal inspiration.

Visualizing lung volumes and capacities...

Importance of FEV1/FVC Ratio

- The proportion of air exhaled in the first second (FEV1) compared to the total volume of air exhaled (FVC).

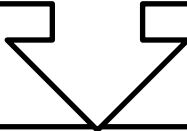
Clinical Significance:

- Important for assessing possible airway obstruction.
- A critical ratio for diagnosing respiratory conditions.

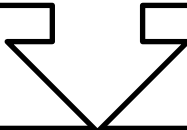
Process	FEV ₁	FVC	FEV ₁ /FVC
Obstructive	<80% of predicted	Reduced, but not to the same degree as FEV ₁	<0.7
Restrictive	<80% of predicted	<80% of predicted	≥0.7

Obstructive Disease

Definition: In obstructive disease, FEV1 is reduced due to increased resistance during expiration.

A large, hollow, downward-pointing arrow connects the definition box to the key feature box.

Key Feature: Air trapping occurs, leading to more air inspired than expired, which increases residual volume.

A large, hollow, downward-pointing arrow connects the key feature box to the diagnosis box.

Diagnosis: FEV1/FVC ratio recovers after administering a bronchodilator (e.g., salbutamol).

Obstructive Disease

Flow-Volume Curve in Obstructive Disease

Observation: "Spoonining" of the flow-volume curve occurs.

Mechanism: Affected small airways begin to collapse.

As air exits, pressure in small airways reduces, increasing resistance to expiration and reducing flow.

Obstructive Disease

**Examples of
Obstructive Diseases
Conditions:**

Asthma

COPD (Chronic
Bronchitis,
Emphysema)

Tracheal Stenosis

Large Airway Tumors

Restrictive Disease

Definition: In restrictive disease, FVC is reduced due to poor lung expansion.

Causes: Neurological issues (weak inspiratory muscles)
Anatomical deformities

Effects: Inspiratory reserve volume is reduced due to limited lung inflation during maximum inspiration.
Residual volume can also be reduced as expiration is typically more effective than inspiration.

Restrictive Disease...

**Examples of
Restrictive Diseases
Conditions:**

Interstitial Pulmonary
Fibrosis

Muscle Weakness

Kyphoscoliosis

Obesity
Tense Ascites

Lung volume and capacities amount

1. The tidal volume :500 milliliters in the adult male.
 2. The inspiratory reserve volume 3000 milliliters.
 3. The expiratory reserve volume 1100 milliliters.
 4. The residual volume is about 1200 milliliters.
1. The inspiratory capacity about 3500 milliliters
 2. The functional residual capacity equals to 2300 milliliters.
 3. The vital capacity equals about 4600 milliliters.
 4. 4. The total lung capacity is equal to 5800

Reference, Guyton and Hall

✓ Session 2:

Physical principles of gas exchange,
and transport.

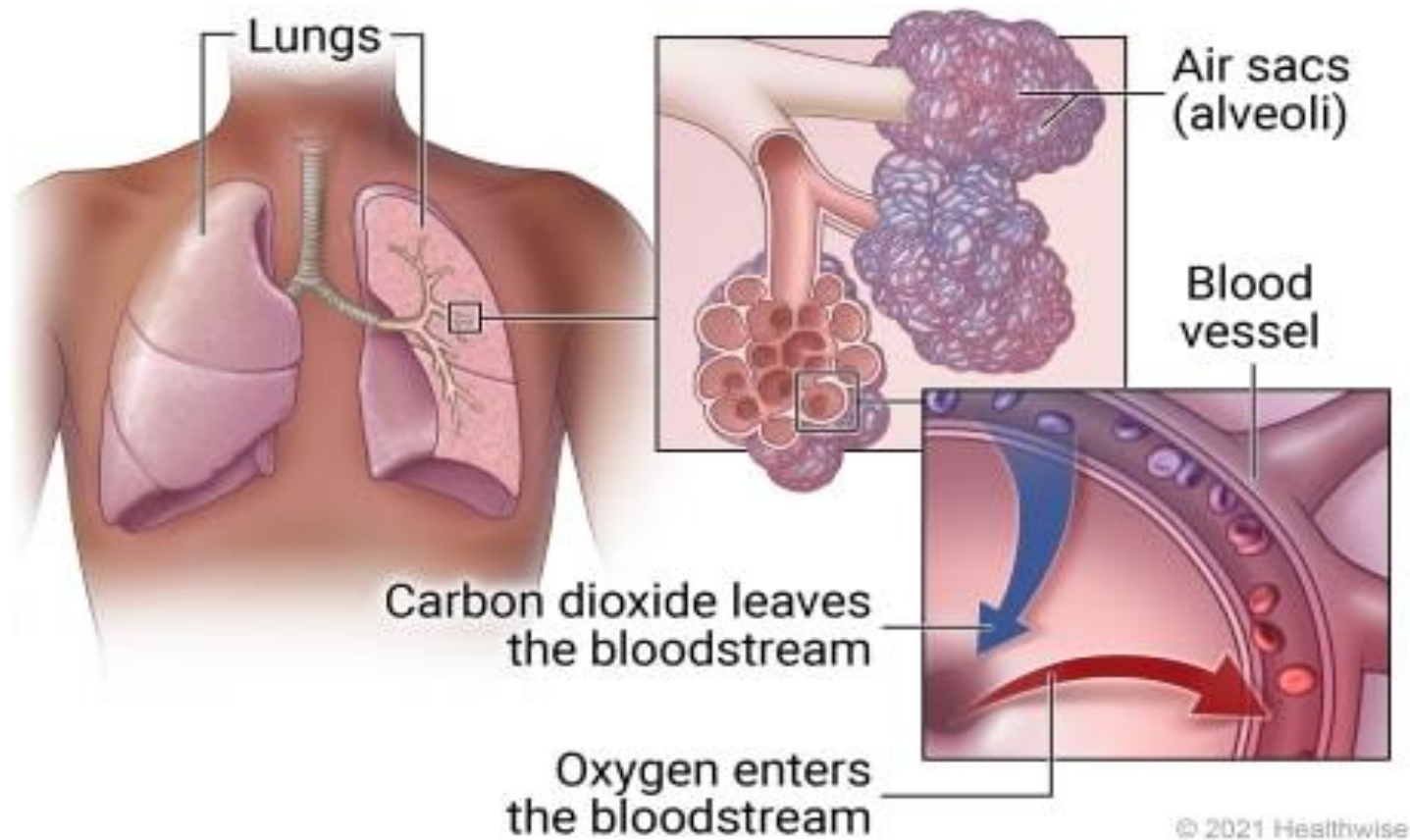
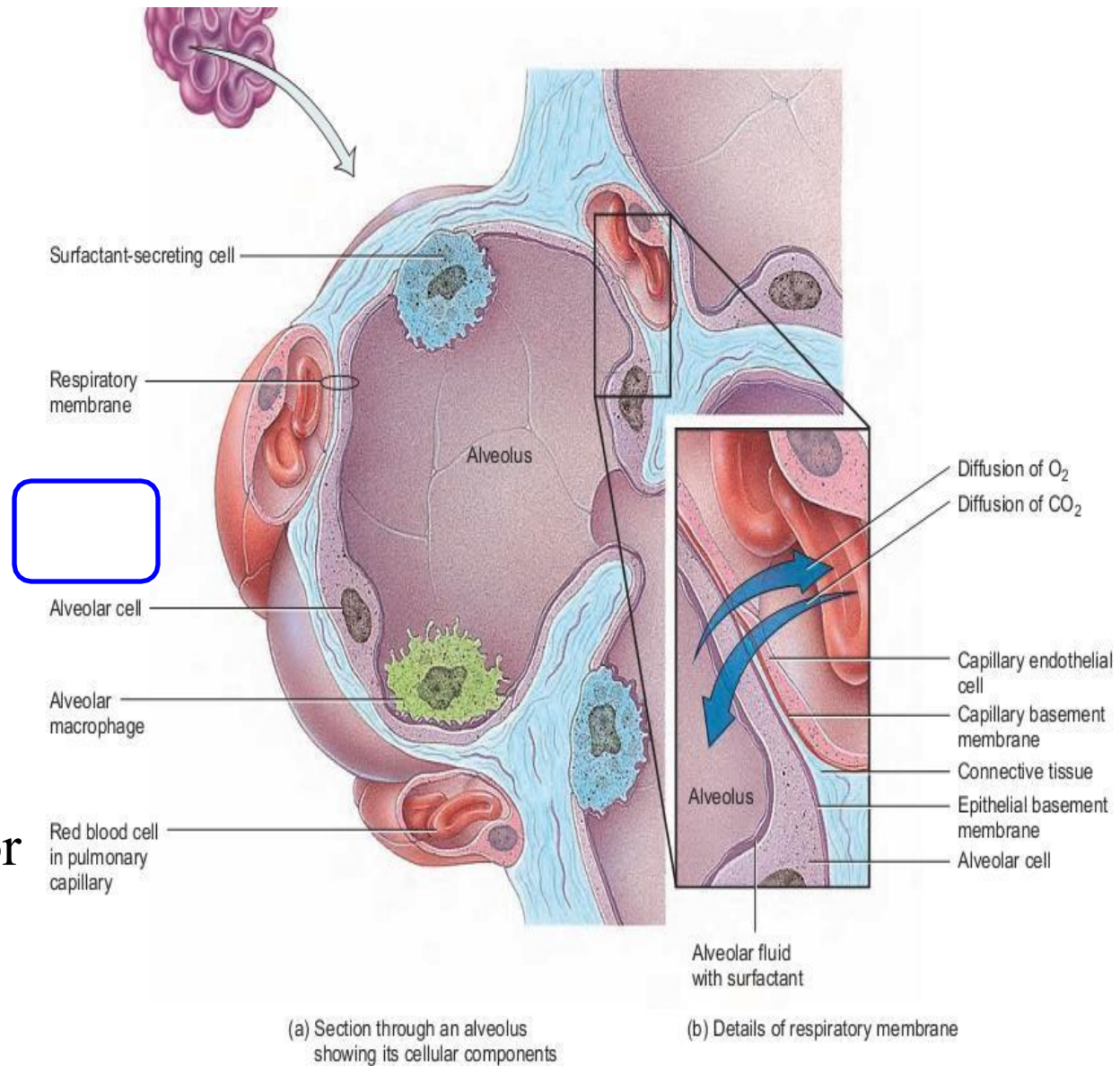


Fig : Gas exchange

Respiratory Membrane

- Air-blood barrier.
- Alveolo-capillary membrane.
- Composed of alveolar wall, capillary wall & their fused basal laminas.
- Exchange of O_2 for CO_2 occurs.



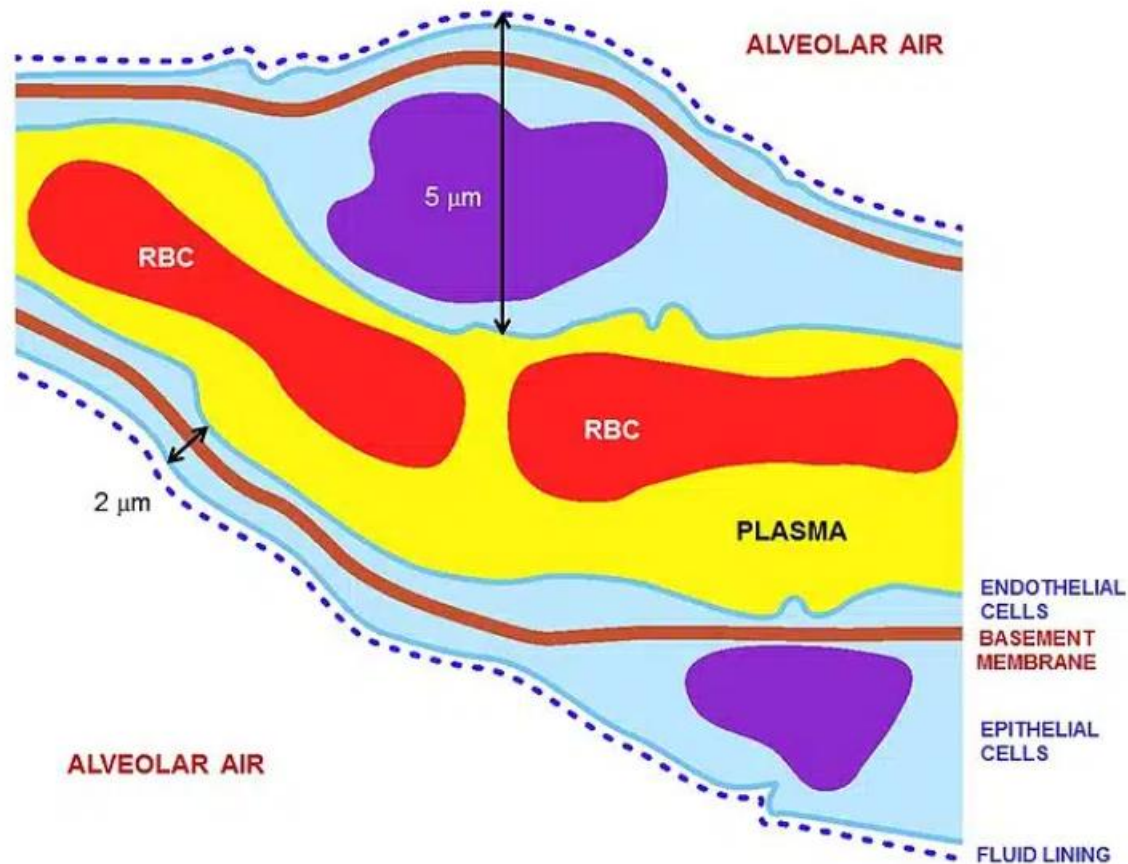


Fig: Diagram showing the layers making up the diffusion barrier in the lungs

Respiratory Membrane ...

□ Alveolar wall is made up of 3 cell types:

1. Type I pneumocytes

- ✓ Permit gas exchange by simple diffusion & secrete angiotensin converting enzyme (**ACE**).

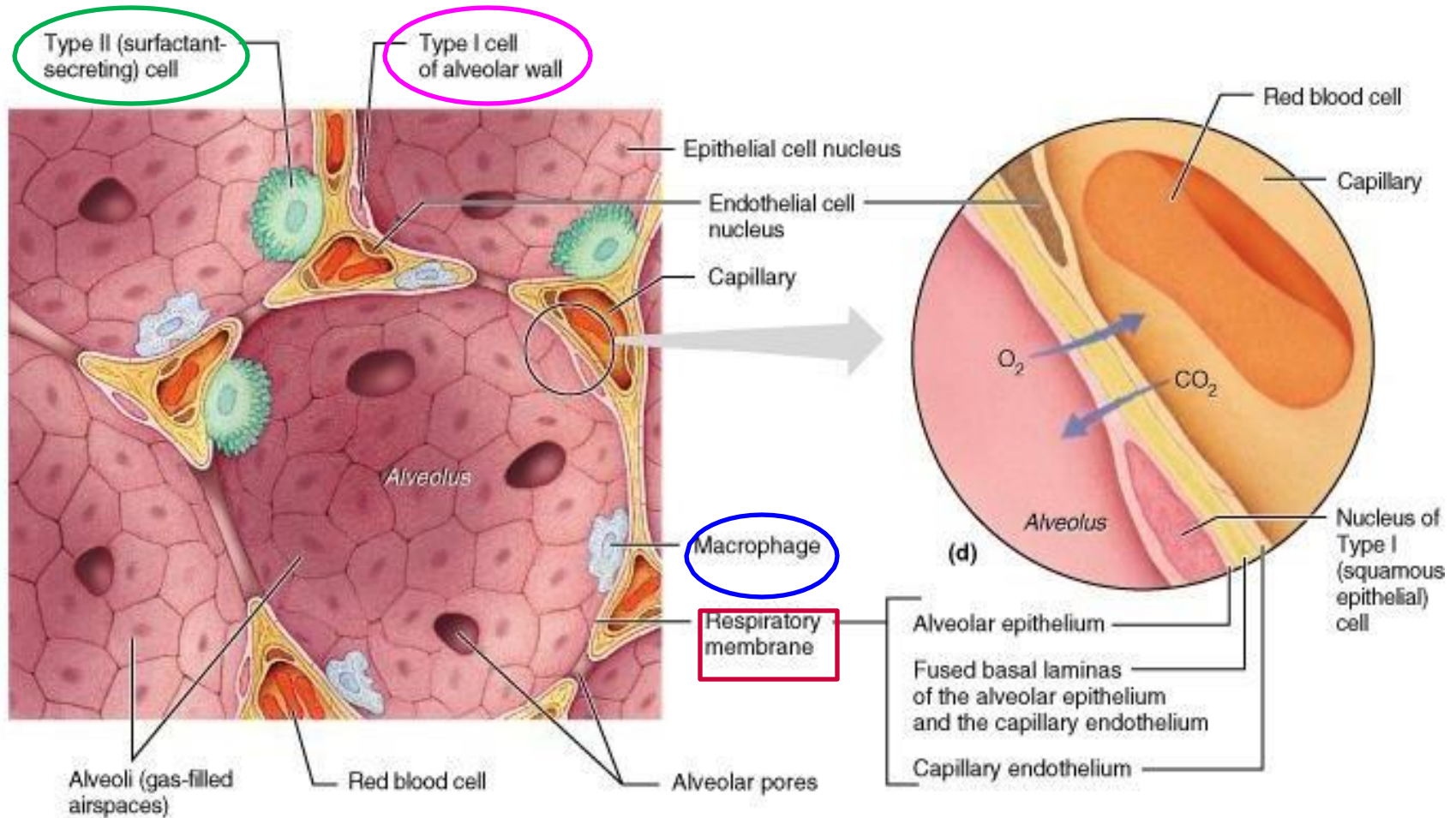
2. Type II pneumocytes

- ✓ Secrete **surfactant** that are less in number & constitute thicker granulocytes in alveolar walls.

3. Type III pneumocytes

- ✓ Alveolar cavity contains large phagocytic cells called **macrophages**.

Respiratory Membrane...



(c)

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Surfactant

- Produced by **type II pneumocytes**.
- Formation of a molecular layer at interface between the liquid lining alveoli & air in the alveoli modifies surface tension → **reduce surface tension**.
 - ↑ distensibility of lungs
 - Reduced recoil force of lungs
 - Prevention of alveolar collapse by balancing pressure between small and large alveoli
 - Reduction in the work of breathing

Blood gas transport & internal respiration

Diffusion of gases:

- O_2 & CO_2 move passively from a region of higher to one of lower concentration.
- Fick's law, the rate of transfer of a gas through a sheet of tissue is:

1. Directly Proportional to:

- tissue surface area(A).
- partial pressure difference between the 2 sides(ΔP).
- fluid temperature(T).
- solubility coefficient of the gas(α).

Blood gas transport...

2. Inversely proportional to:

- ✓ fluid viscosity(η).
- ✓ tissue thickness(L).
- ✓ square root of molecular weight of the gas($\sqrt{MW}$).

Blood gas transport...

■ Factors affecting diffusion capacity of lungs (DL)

A) Properties of gases

- Solubility coefficient of gas, absolute temperature of fluid, $\sqrt{\text{MW}}$ of gas

B) Lung properties

1. Thickness of path length of membrane
2. Size of alveolar surface area
3. Permeability of membrane
4. Fluid viscosity in lung
5. Pulmonary capillary blood volume

C) Other factor

Partial pressure difference b/n alveoli & capillaries .

$$\propto \frac{\Delta P T A \alpha}{\eta L \sqrt{\text{MW}}}$$

Formula

O₂ & CO₂ diffusion gradients

O₂

- Moves **from alveoli** into blood.
- Blood is almost completely saturated with O₂ when it leaves the capillary.
- P_{O₂} in blood ↓ because of **mixing** with deoxygenated blood.
- O₂ moves from tissue capillaries into the tissues.

CO₂

- Moves **from tissues** into tissue capillaries.
- Moves from pulmonary capillaries into the alveoli.

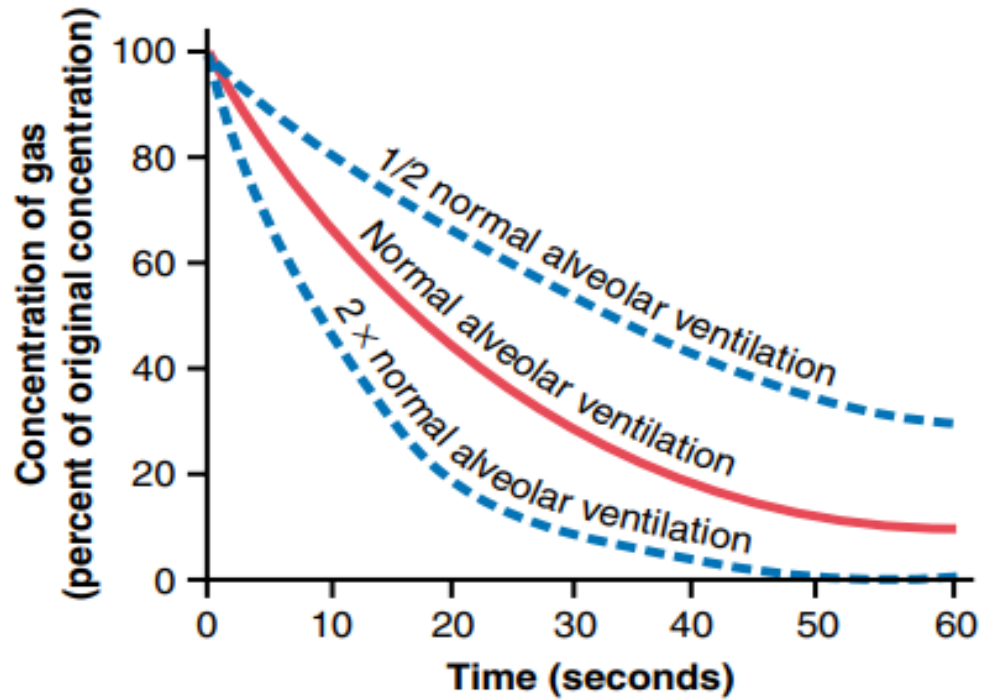


Fig: Rate of removal of excess gas from alveoli.

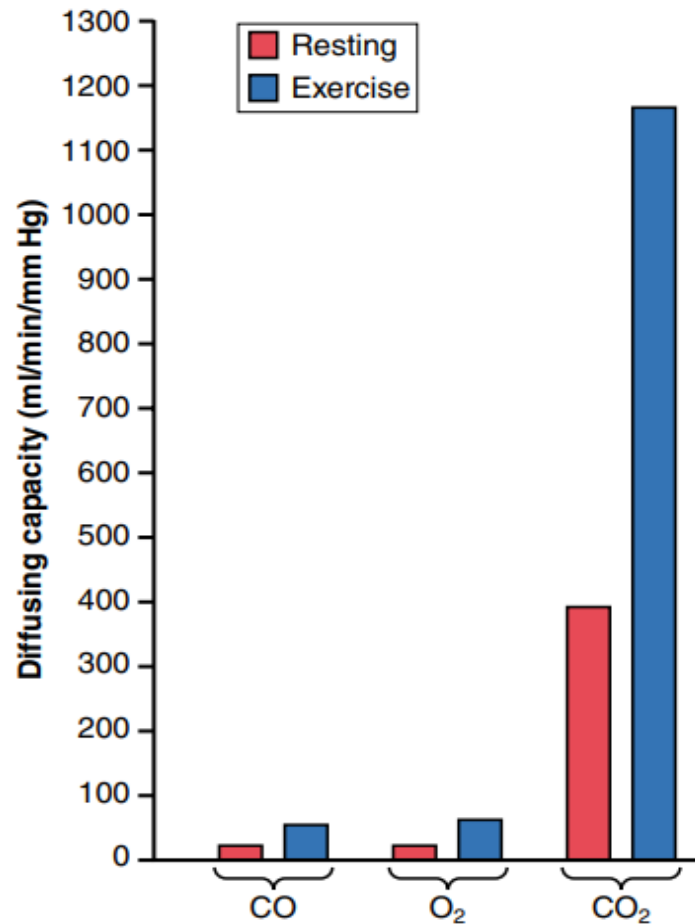
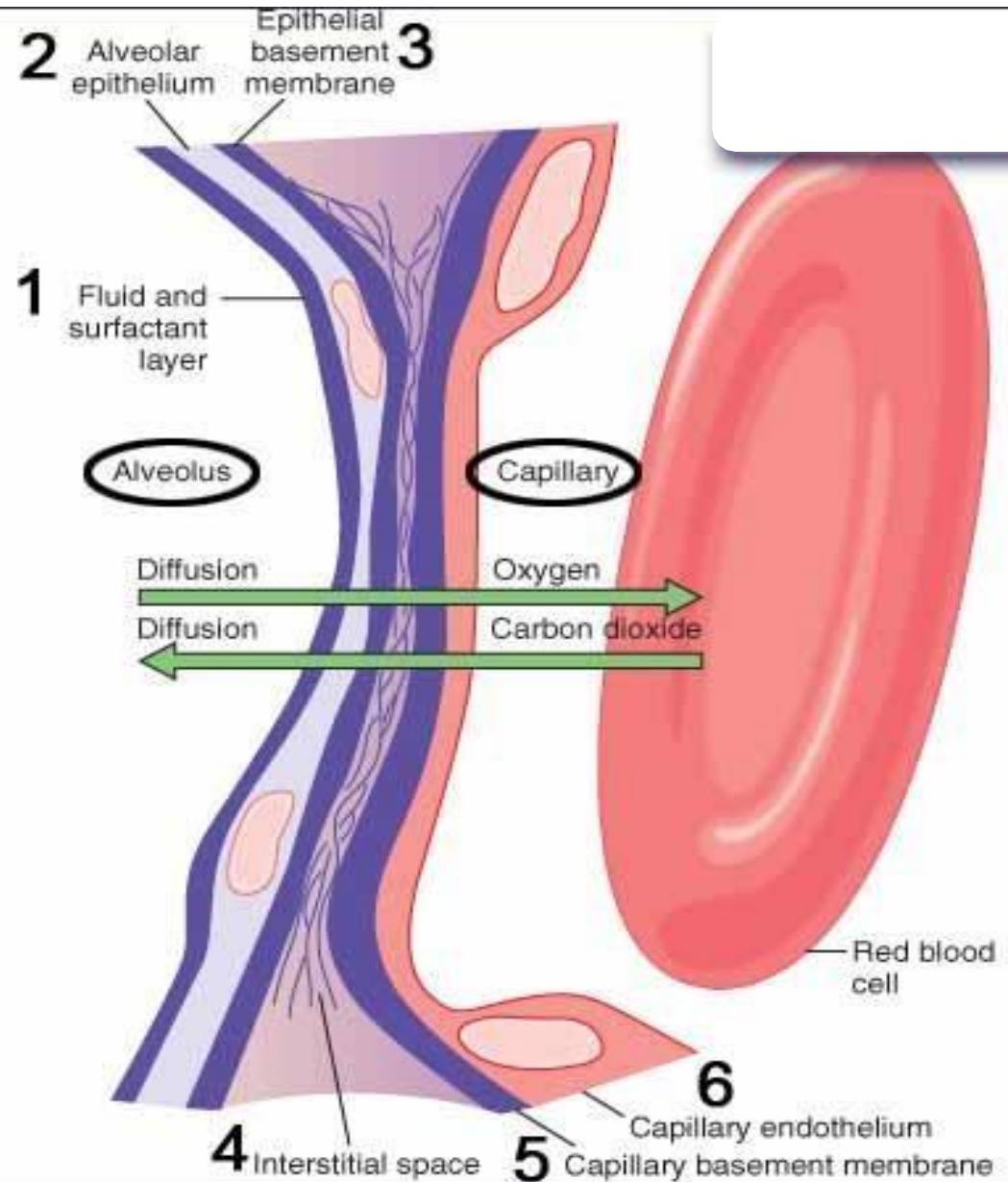


Fig : Diffusing capacities for carbon monoxide, oxygen, and carbon dioxide in the normal lungs under resting conditions and during exercise

O₂ & CO₂
diffusion
gradients...



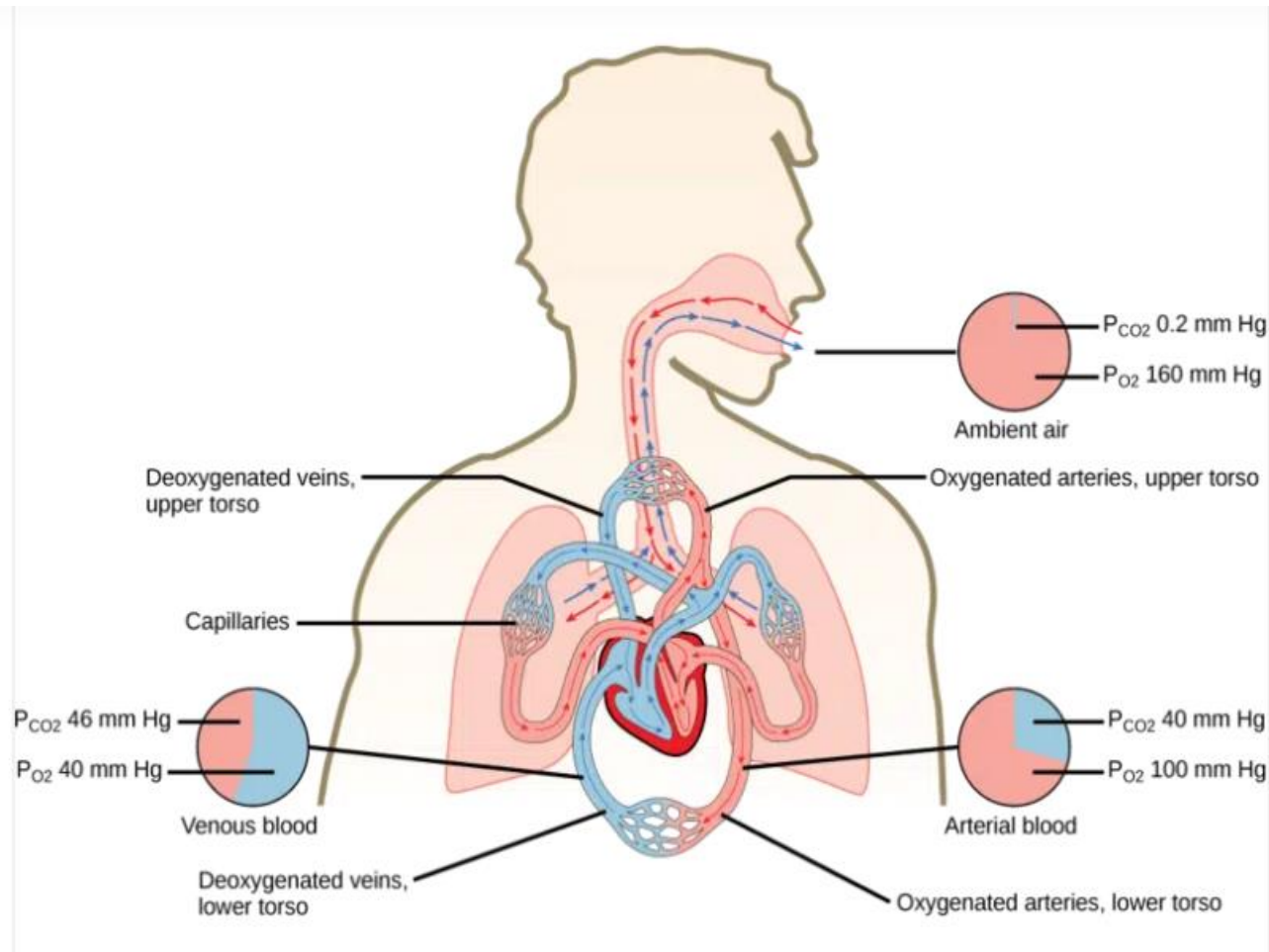
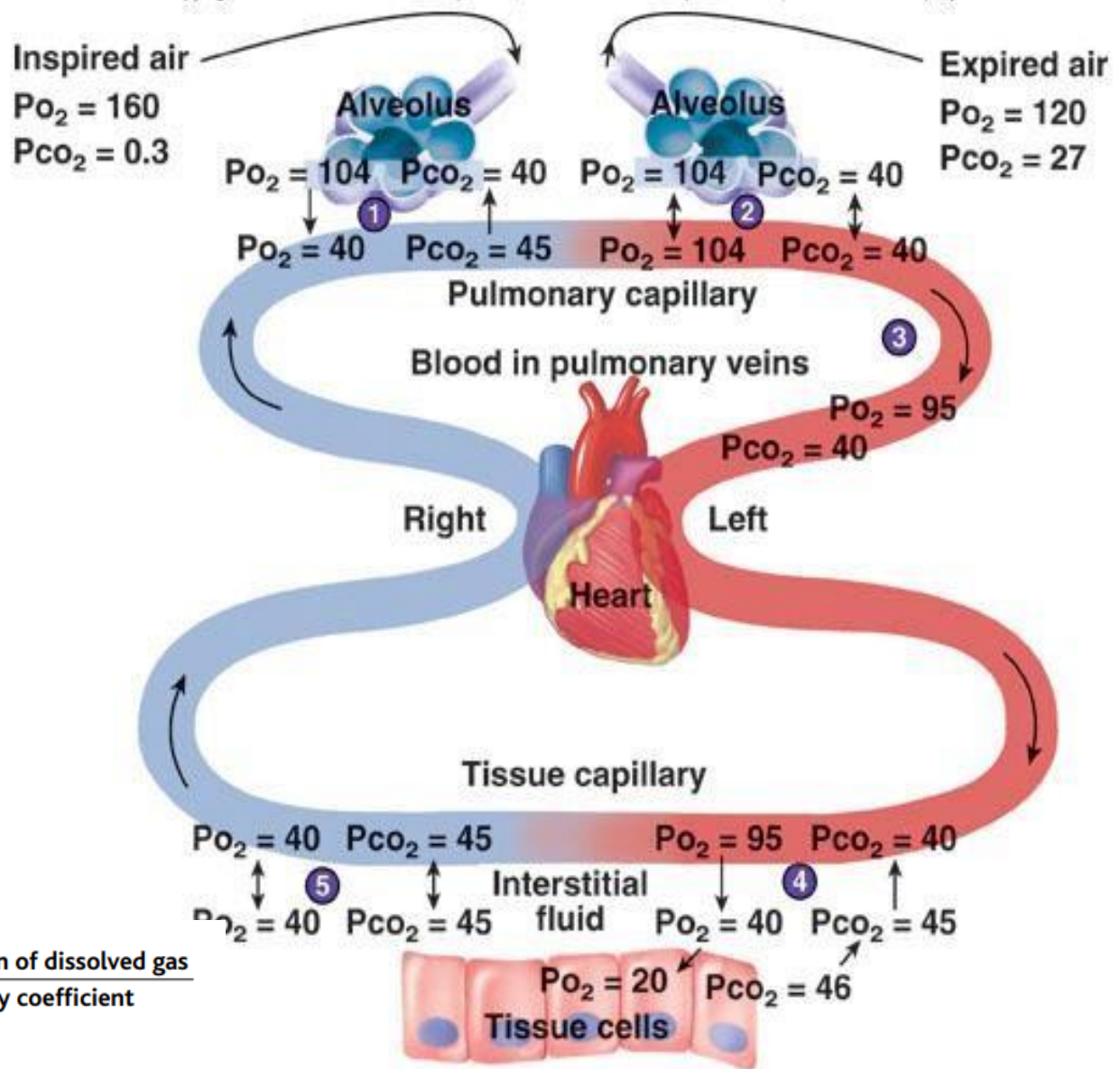


Fig: Diagram showing the partial pressures of oxygen and carbon dioxide in the respiratory system



Blood gas transport

...

$$\text{Partial pressure} = \frac{\text{Concentration of dissolved gas}}{\text{Solubility coefficient}}$$

Oxygen Transport

- **Oxygen** is transported by:
 - ✓ **Hemoglobin** (97%) and
 - ✓ Dissolved form in **plasma** (3%)
- **O₂-hemoglobin dissociation curve** shows that hemoglobin is almost completely saturated when **P_{O₂}** is **80 mmHg** or above.
- At lower partial pressures, hemoglobin releases O₂.
- The substance 2,3-bisphosphoglycerate ↑ the ability of **hemoglobin to release O₂**.

Oxygen
Hemoglobin
dissociation
curve
at
rest

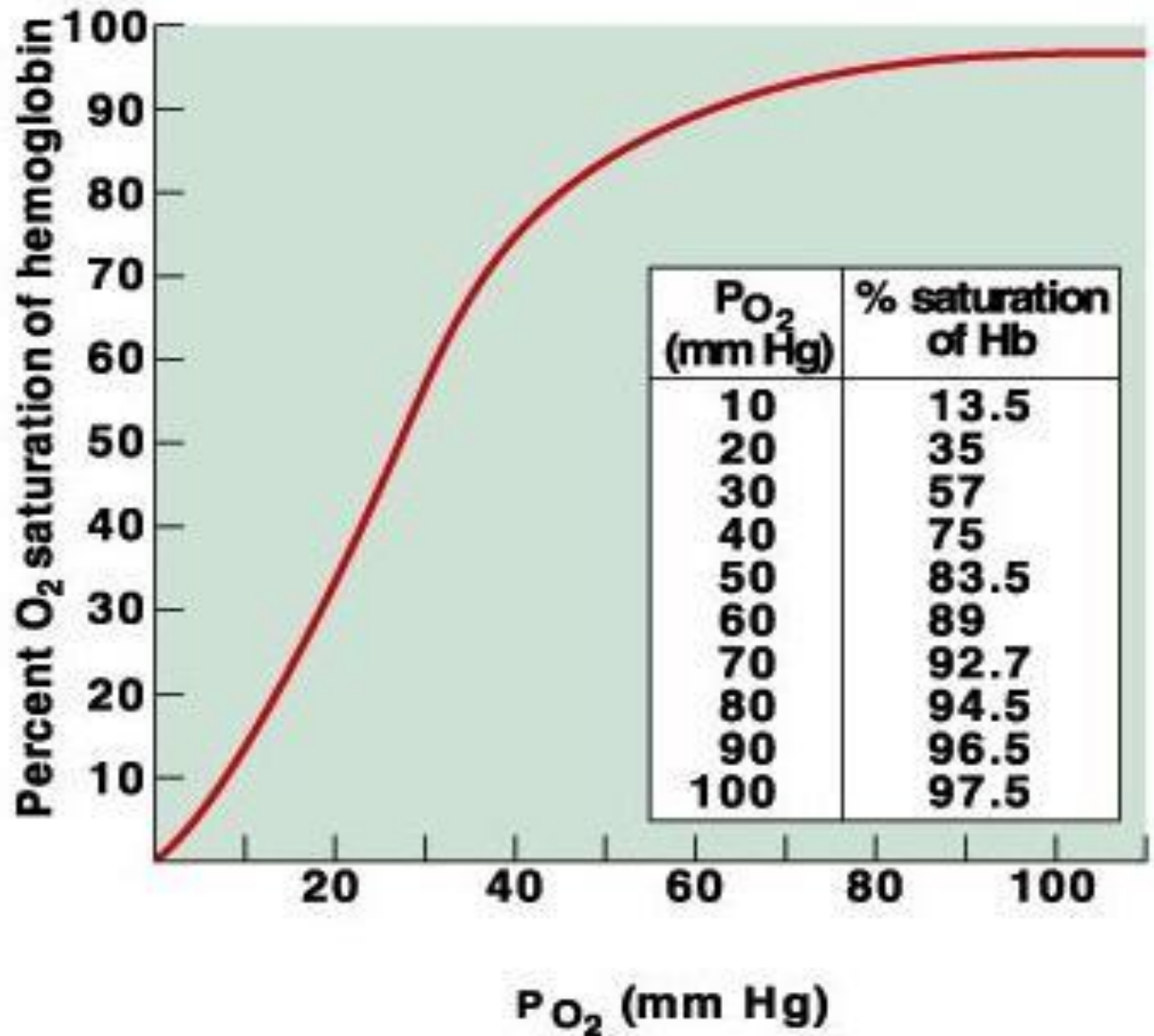
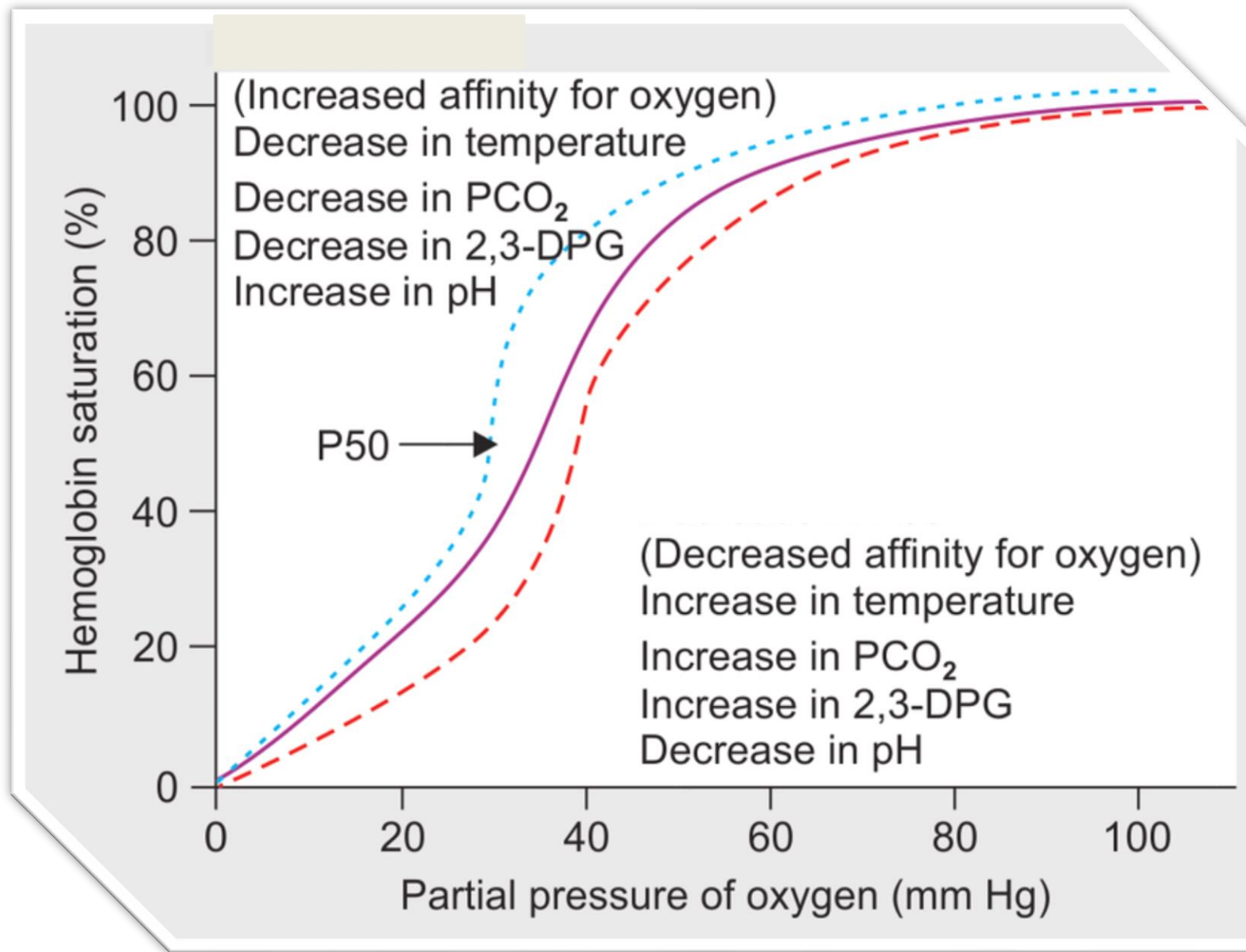


Fig:
Causes of
Shift of the
curve to
Right or
left.



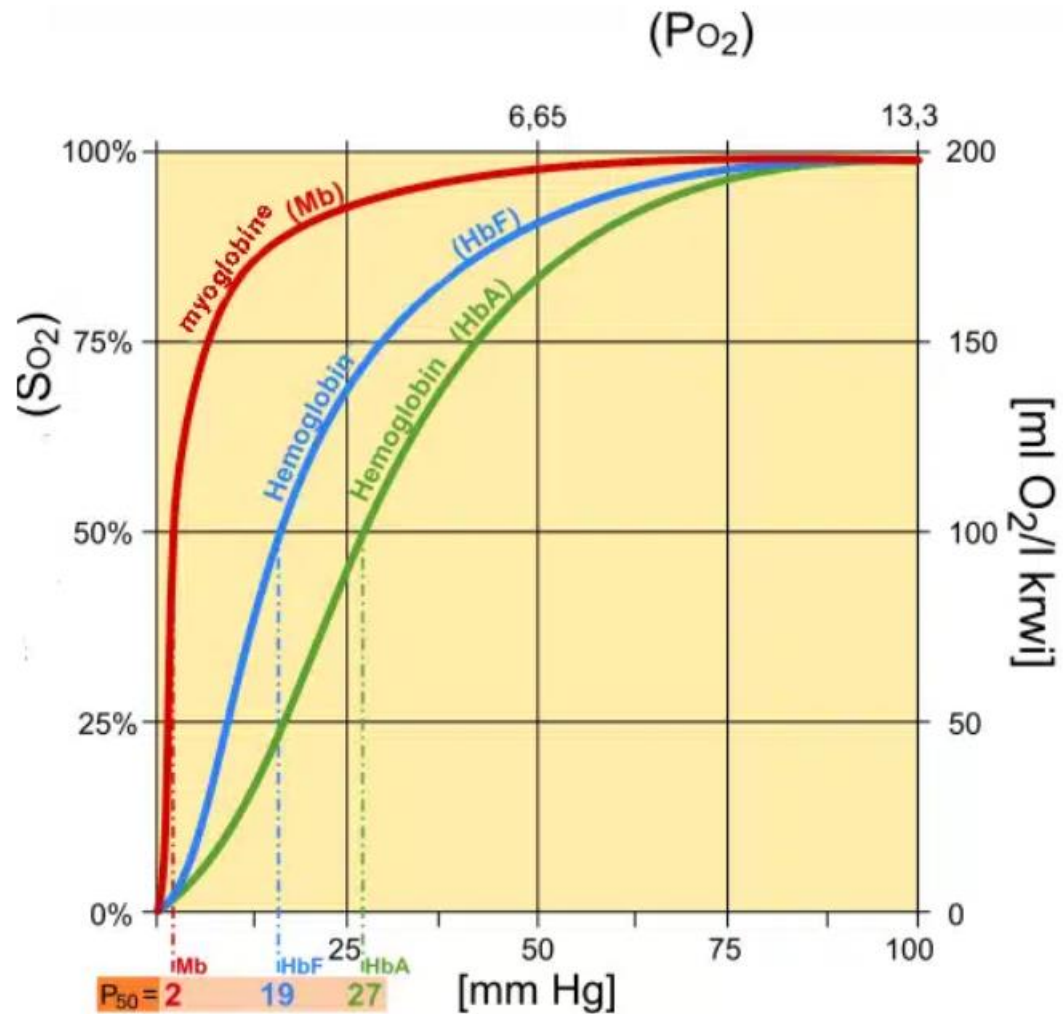


Fig: Oxygen dissociation curve showing the relative properties of myoglobin, HbF, and HbA

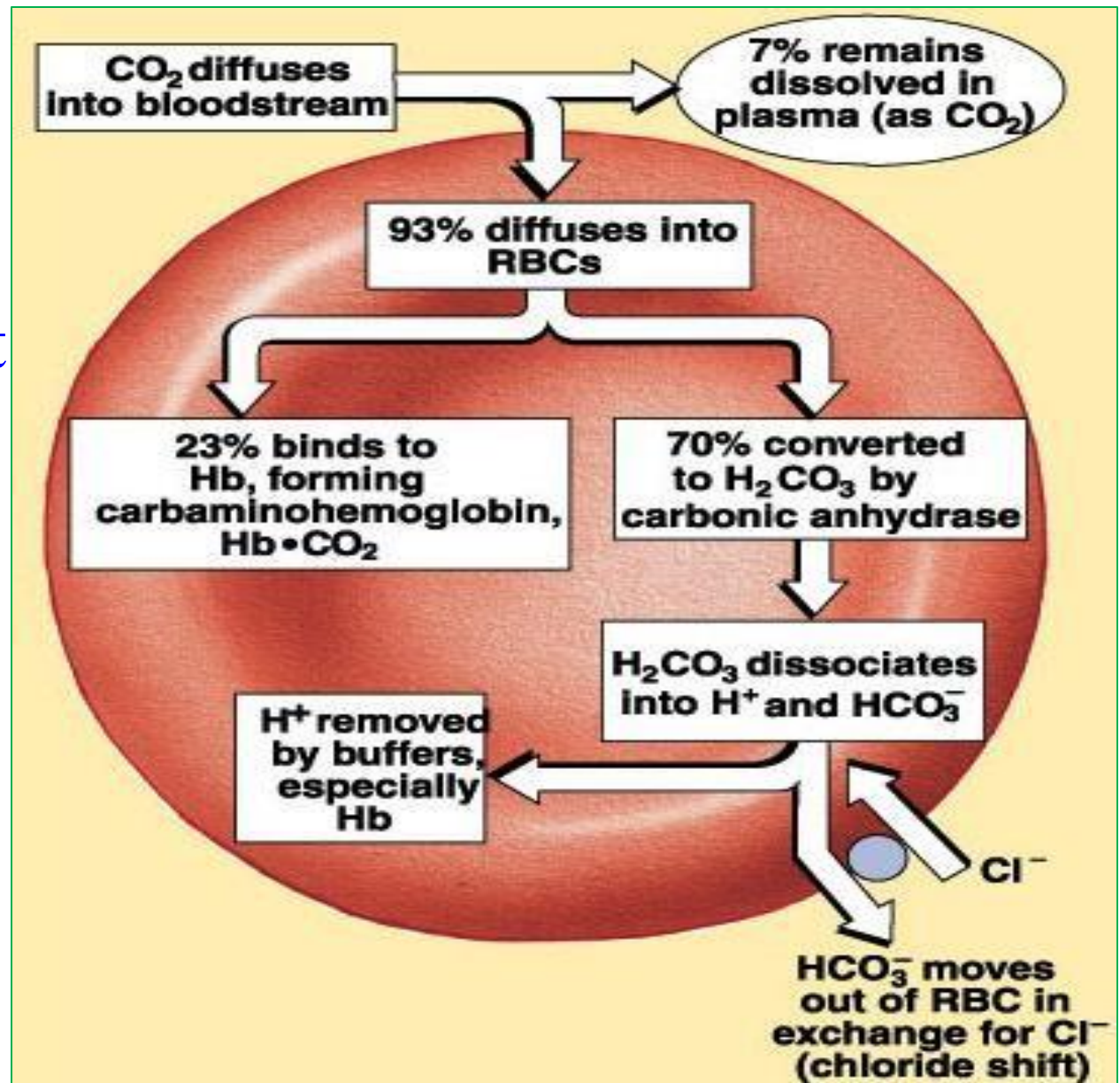
Oxygen Transport...

- ❑ A shift of the curve to the **left** because of
 - ✓ an $\uparrow$ in **PH**,
 - ✓ a $\downarrow$ in **PCO₂** or a $\downarrow$ in **temperature**
 - results in an $\uparrow$ in the ability of **hemoglobin** to hold O₂.
- ❑ A shift of the curve to the **right** because of
 - ✓ a $\downarrow$ in **PH**,
 - ✓ an $\uparrow$ in **PCO₂** or an $\uparrow$ in **temperature**
 - results in a $\downarrow$ in the ability of **hemoglobin** to hold O₂.

Transport of Carbon Dioxide

- **Carbon dioxide** is transported in the form of;
 1. **Bicarbonate** -70%
 2. **Carbaminohemoglobin** - 23%
 3. Dissolved in **plasma** - 7%
- **Haldane effect**
 - ✓ Hemoglobin that has released O_2 binds more readily to CO_2 than hemoglobin that has O_2 bound to it.

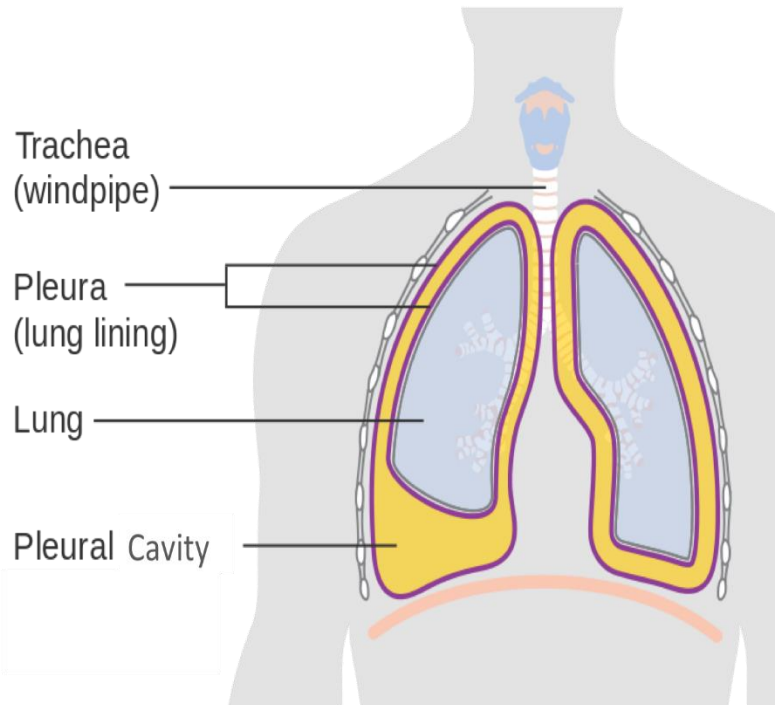
CO₂ transport & Cl⁻ movement



✓ Session 3:

Pulmonary Circulation ,edema and
pleural fluids.

Pleural fluid



The space between the outer surface of the lungs and inner thoracic wall is known as the **pleural space**.

This is usually filled with **pleural fluid**, forming a seal which holds the lungs against the thoracic wall by the force of surface tension.

This seal ensures that when the thoracic cavity expands or reduces, the lungs undergo expansion or reduction in size accordingly.

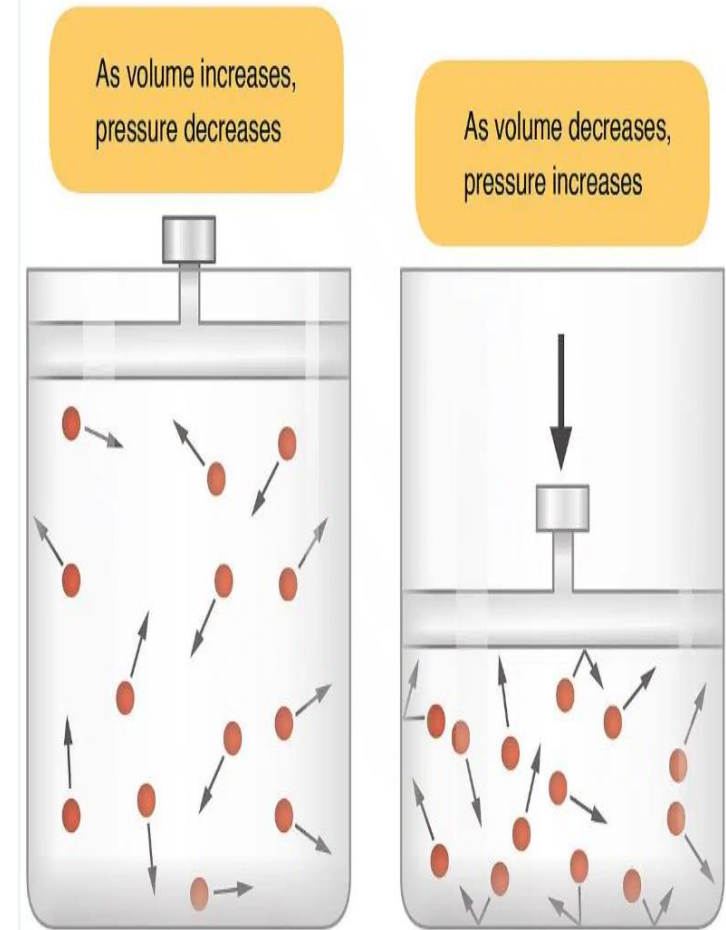
During breathing, the contraction and relaxation of muscles acts to change the volume of the thoracic cavity.

As the thoracic cavity and lungs move together, this changes the **volume** of the lungs, in turn changing the **pressure** inside the lungs.

Pleural fluid...

Boyle's law states that the volume of gas is inversely proportional to pressure (when temperature is constant). Therefore:

- **When the volume of the thoracic cavity increases** – the volume of the lungs increases and the pressure within the lungs decreases.
- **When the volume of the thoracic cavity decreases** – the volume of the lungs decreases and the pressure within the lungs increases.



Two circulations in the respiratory system

- **Bronchial Circulation**

- Arises from the aorta.
 - Part of systemic circulation (oxygenated).
 - Receives about 1-2% of left ventricular output.
 - Supplies the supporting tissues of the lungs, including the connective tissue, septa, and bronchi.
 - It empties into the pulmonary veins and eventually into left atrium
-

Pulmonary blood flow

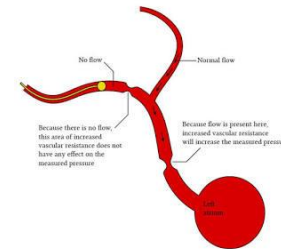
- Pulmonary Pressures

- Pulmonary artery pressure

- systolic 25 mmHg
 - diastolic 8 mmHg
 - mean 15 mmHg
 - capillary 10 mmHg

- Left Atrial and Pulmonary Venous Pressures = 2 (1-5) mm Hg (estimated)
- Pulmonary wedge pressure = 5 mm Hg

(usually its 2 to 3 mm Hg greater than the left atrial pressure)



Why Pressures Are Different in Pulmonary and Systemic Circulations ?

- Gravity and Distance:
 - Distance above or below the heart adds to, or subtracts from, **both** arterial and venous pressure
 - Distance between Apex and Base affected by gravity

Systemic

Aorta 100 mmHg

Head 50 mmHg

Feet 180 mmHg

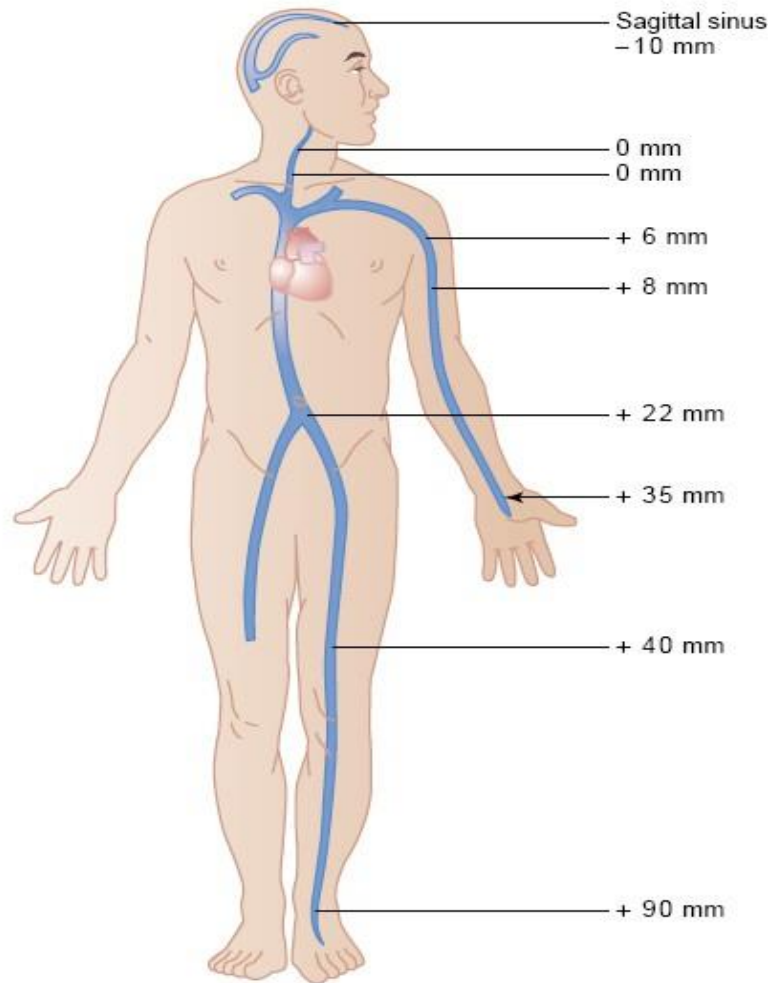
Pulmonary

Mean PA 15 mmHg

Apex 2 mmHg

Base 25 mmHg

Effect of hydrostatic pressure on venous pressure in the standing position



Difference between pulmonary capillary & that of systemic

	Pulmonary capillary	Systemic capillary
P_c : Capillary hydrostatic pressure	10 mm Hg	17 mm Hg
P_c : Capillary oncotic pressure (due to proteins in the blood)	28 mmHg	28 mmHg
P_i : Interstitial hydrostatic pressure.	- 5 mmHg	Zero
P_i : Interstitial oncotic pressure (due to proteins in the interstitial fluid).	14 mmHg	7 mmHg

Pressure in the different areas of the lungs

- At the top, 15 mm Hg less than the pulmonary arterial pressure at the level of the heart
- At the bottom, 8 mm Hg greater than the pulmonary arterial pressure at the level of the heart.
- 23 mm Hg pressure difference between the top and the bottom of the lung
- These differences have effects on blood flow through the different areas of the lungs.

Pulmonary resistance to flow

- Pressure drop of 12 mmHg (P_m -PRA)
- Flow of 5 l/min
- Resistance 1/7 systemic circulation

Pulmonary Capillary Dynamics

- **Outward Forces**

- Pulmonary capillary pressure 10 mmHg
- Interstitial colloid osmotic pressure 14 mmHg
- Negative interstitial pressure 5 mmHg
- **Total 29 mmHg**

- **Inward Forces**

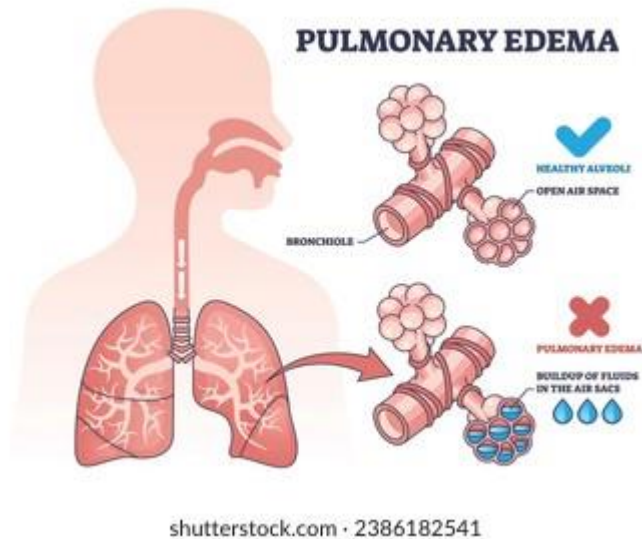
- Plasma osmotic pressure 28 mmHg

- **Net filtration pressure 1 mmHg**

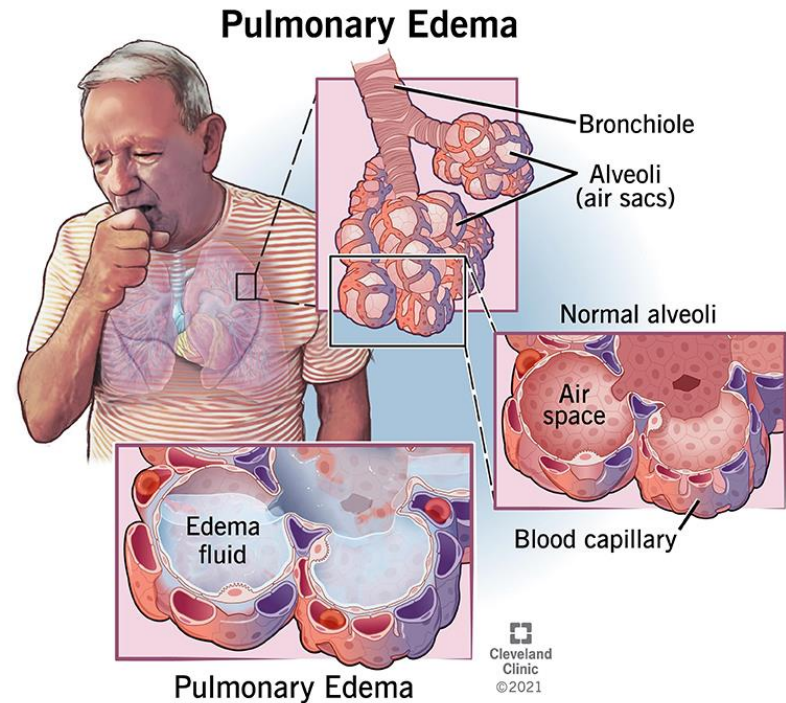
- **Lymphatic vessels take care of this extra filtrate**

- There is plenty lymphatics which empty in the right lymphatic duct to prevent the occurrence of pulmonary edema.

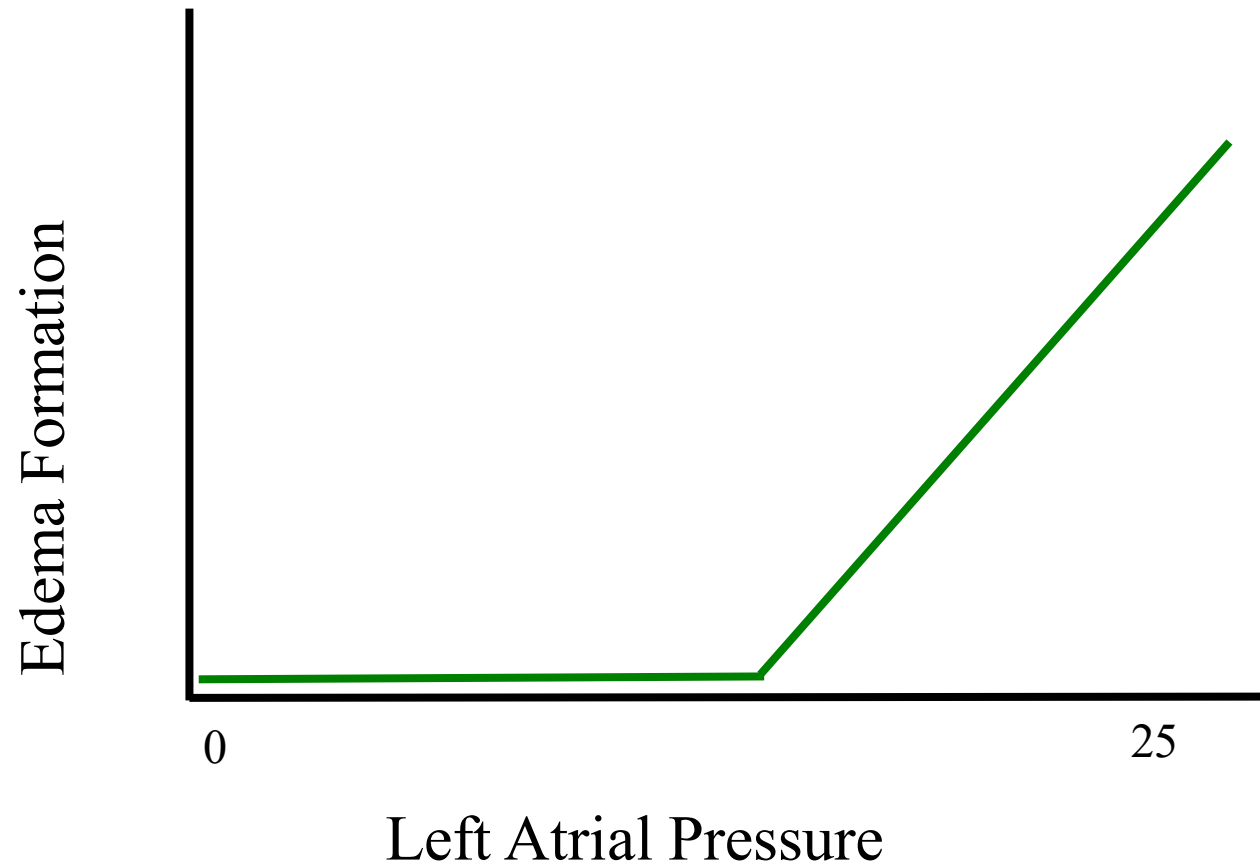
Pulmonary Edema



- Pulmonary edema is a condition where excess fluid builds up in the lungs' air sacs, making it difficult to breathe because oxygen cannot enter the bloodstream.



Heart Failure and Pulmonary Edema



Pulmonary Edema causes

Pulmonary oedema - causes

Condition in which fluid accumulates within the alveolar space resulting in impaired pulmonary gas exchange

Usually results from increased hydrostatic pressure in the pulmonary venous circuit

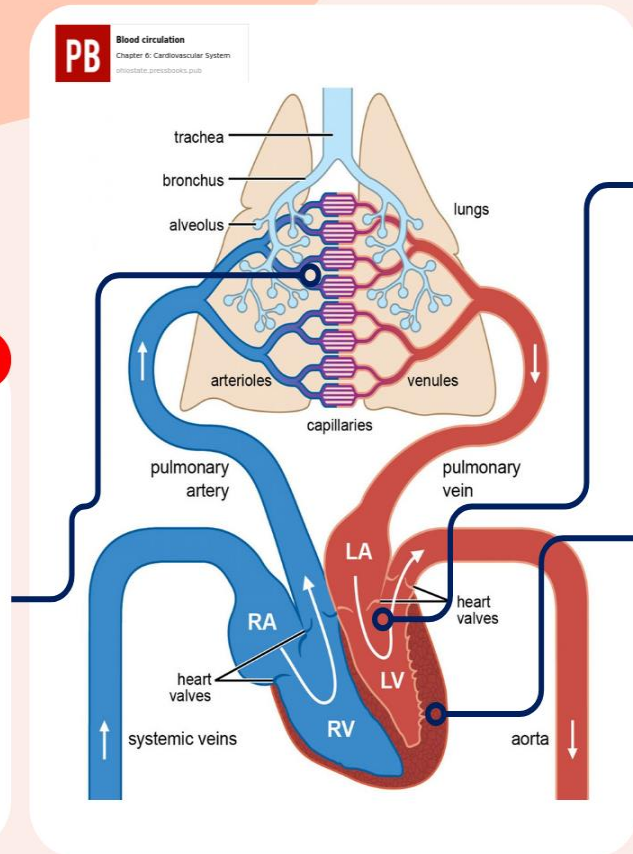
Non-cardiogenic

Damage to lung parenchyma causes increased capillary permeability and escape of fluids and proteins into alveolar space

ALI/ ARDS

Intravascular volume expansion

Chronic and acute kidney injury
Transfusion overload
Severe hypoproteinaemia



Left valvular dysfunction

Obstruction of blood flow from LA to LV leads to back-pressure to PV and increased pulmonary venous and capillary hydrostatic pressure

Mitral stenosis/ regurgitation
Aortic stenosis

Left ventricular failure (LVF)

Reduced ejection fraction (EF) causes increased pre-load, pooling of blood in cardiac chambers and back-pressure to PV with increased pulmonary venous and capillary hydrostatic pressure

Myocardial infarction
Cardiomyopathies, myocarditis
Cardiac arrhythmias (AF, VT, HB)

Pulmonary Blood Flow

- The alveolar capillaries are distended by the blood pressure inside them and compressed by the alveolar air pressure on their outsides.
- If the alveolar air pressure (P_{alv}) becomes greater than the pulmonary capillary blood pressure (P_{pc}), the capillaries will close and there is no blood flow.

Measurement of pulmonary blood flow

$$V_{O_2} = Q(Ca_{O_2} - Cv_{O_2})$$

V_{O_2} = Oxygen Consumption

Ca_{O_2} = Arterial Content

Q = Blood flow

Cv_{O_2} = Venous Content

Measurement of pulmonary blood flow

$$V_{O_2} = Q(Ca_{O_2} - Cv_{O_2})$$

$$V_{O_2} = 250 \text{ ml/min}$$

Q?

$$Ca_{O_2} = 20 \text{ ml O}_2/100 \text{ ml blood}$$

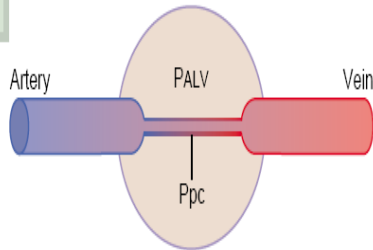
$$Cv_{O_2} = 15 \text{ ml O}_2/100 \text{ ml blood}$$

$$Q = \frac{250 \text{ ml O}_2/\text{min}}{(20-15) \text{ ml O}_2/100 \text{ ml blood}} = \frac{250 \text{ ml O}_2}{\text{min}} * \frac{100 \text{ ml blood}}{5 \text{ ml O}_2}$$

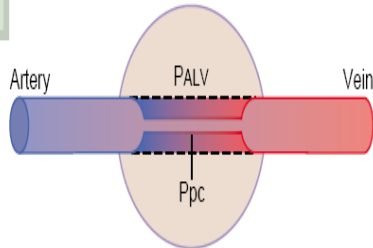
$$Q = 5000 \text{ ml blood /min}$$

Zones of Pulmonary Blood Flow

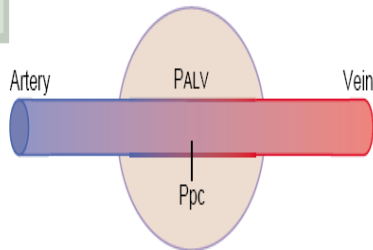
ZONE 1



ZONE 2



ZONE 3



- **Zone 1:**

- ✓ **no flow**

- ✓ alveolar air pressure (Palv) is higher than pulmonary arterial pressure (Ppc) during any part of cardiac cycle... This zone does not exist in human lung.

- **Zone 2:**

- ✓ **intermittent flow**

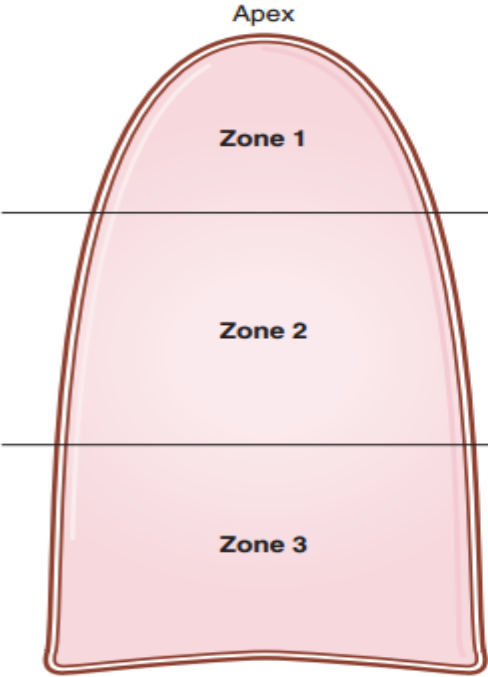
- ✓ systolic arterial pressure higher than alveolar air pressure, but diastolic arterial pressure below alveolar air pressure.

- **Zone 3:**

- ✓ **continuous flow**

- ✓ pulmonary arterial pressure (Ppc) remain higher than alveolar air pressure at all times.

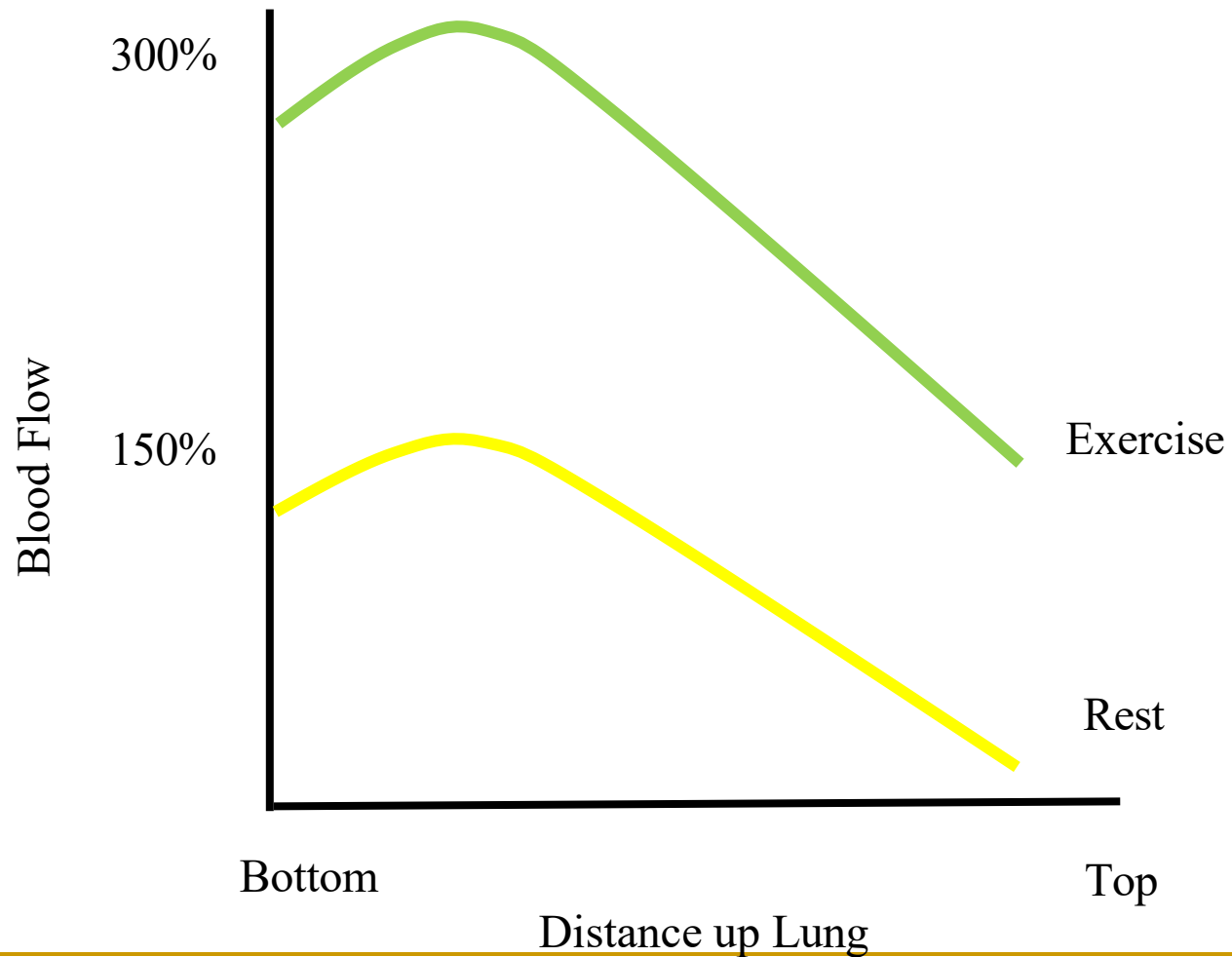
Zones of Pulmonary Blood Flow

BLOOD FLOW DISTRIBUTION IN THE LUNG		
	Blood Flow	Pressures Driving Blood Flow
Apex  Zone 1	Lowest	$P_A \geq P_a > P_v$
Zone 2	Medium	$P_a > P_A > P_v$
Zone 3 Base	Highest	$P_a > P_v > P_A$

Zones of a normal lung

- Normally, the lungs have 2 zones for blood flow
 - zone 2 (intermittent flow) at the apices.
 - zone 3 (continuous flow) in all the lower areas.
 - In normal lungs, Zone 2 begins 10 cm above the mid- level of the heart to the top of the lungs.
-

Distribution of blood flow



Pulmonary Circulation

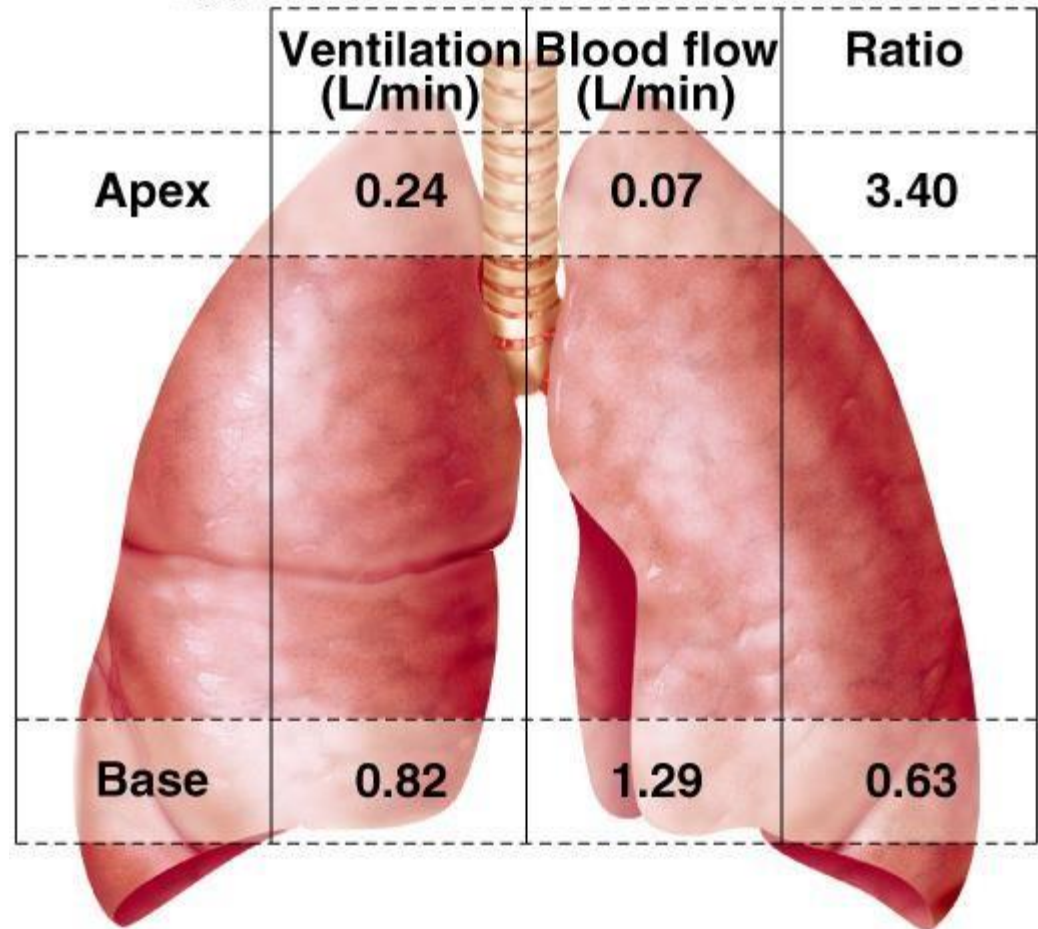
- Rate of blood flow through the pulmonary circulation is = rate of flow through the systemic circulation.
 - Driving pressure in pulmonary circulation is only 10 mm Hg.
- Pulmonary vascular resistance is low.
 - Low pressure pathway produces less net filtration than produced in the systemic capillaries.
 - Avoids pulmonary edema.
- Autoregulation:
 - Pulmonary arterioles constrict when alveolar PO_2 decreases.
 - Matches ventilation/perfusion ratio.

Lung Ventilation/Perfusion Ratios

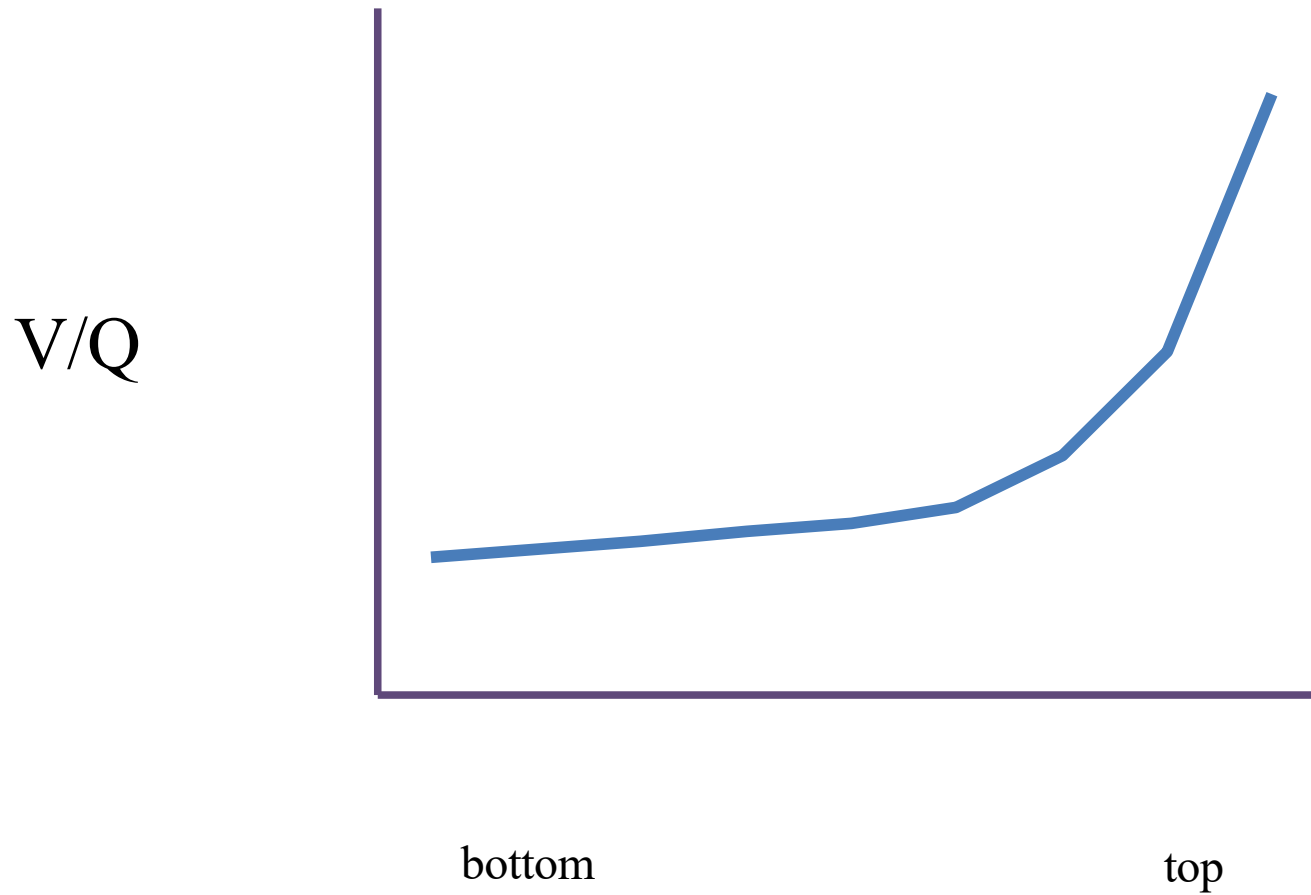
- Functionally:
 - Alveoli at apex are underperfused (overventilated).
 - Alveoli at the base are underventilated (overperfused).

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	Ventilation (L/min)	Blood flow (L/min)	Ratio
Apex	0.24	0.07	3.40
Base	0.82	1.29	0.63

A diagram of human lungs is shown, with a table overlaid on it. The table has four columns: an unlabeled column for location, 'Ventilation (L/min)', 'Blood flow (L/min)', and 'Ratio'. The rows are 'Apex' and 'Base'. The lungs are depicted in a reddish-pink color, with the trachea visible in the center. The table is divided into three horizontal sections by dashed lines, with the 'Apex' and 'Base' labels positioned in the top and bottom sections respectively.

Regional V/Q Ratio



V/Q Ratio

- V/Q is $\uparrow$ in:
 1. pulmonary embolism.
 2. emphysema.
 3. cigarette smokers.
 4. pulmonary hyperventilation
- Whenever V/Q $\uparrow$
 1. alveolar dead space $\uparrow$.
 2. mixed expired $P_E\text{CO}_2 \downarrow$.
 3. mixed expired $P_E\text{O}_2 \uparrow$.

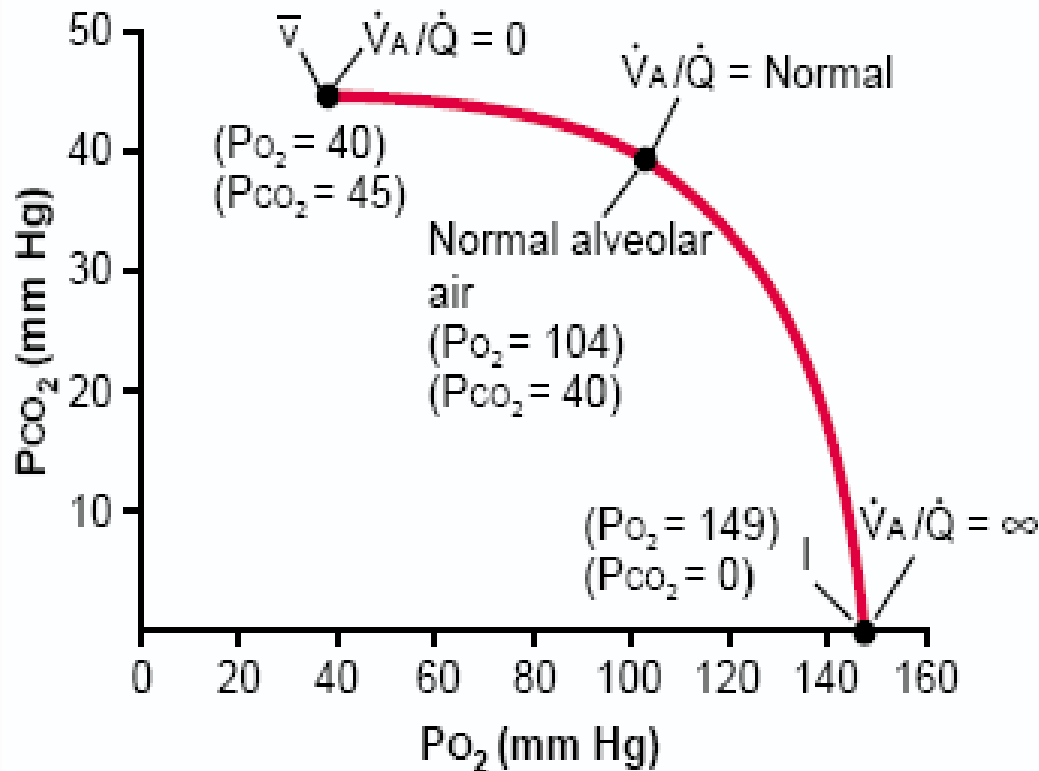
Ventilation/perfusion

- Relationship between adequate flow and adequate ventilation is the V/Q ratio
- $V/Q = (4.2 \text{ l/min}) / (5 \text{ l/min}) = 0.84$
- If there is no diffusion impairment, then the PO_2 and PCO_2 between an alveolus and end capillary blood are usually the same. $P_AO_2 - P_aO_2 \approx \text{zero}$

Ventilation/perfusion ratio

- The ratio of alveolar ventilation to pulmonary blood flow = **0.84** (4.2 L/min ÷ 5 L/min).
- When the ventilation (V) is zero, but there is adequate perfusion (Q) of the alveolus, the V/Q is **zero**.
- when there is adequate ventilation, but zero perfusion, the ratio V/Q is **infinity**.
- At a ratio of either **zero** or **infinity**, there is **no exchange of gases** through the respiratory membrane of the affected alveoli

Ventilation/perfusion ratio



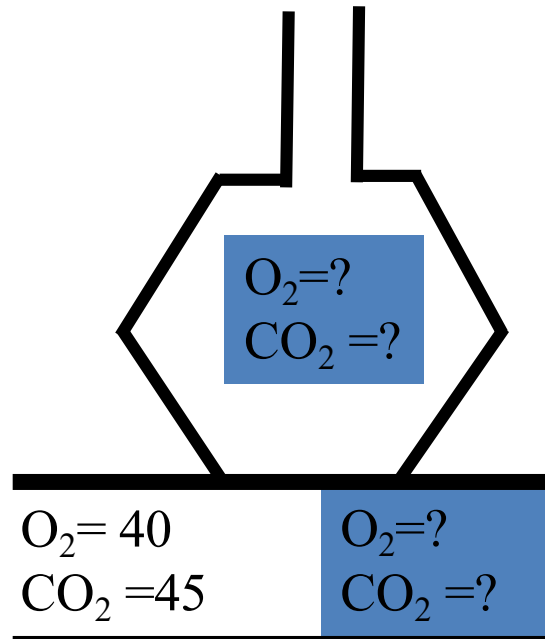
physiologic shunt:

- The total amount of shunted blood per minute.

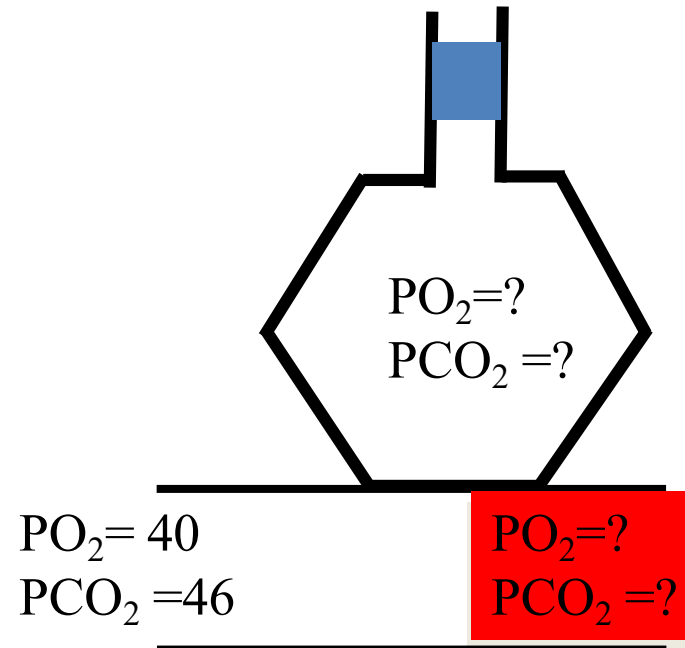
physiologic dead space:

- Alveolar + anatomical dead spaces

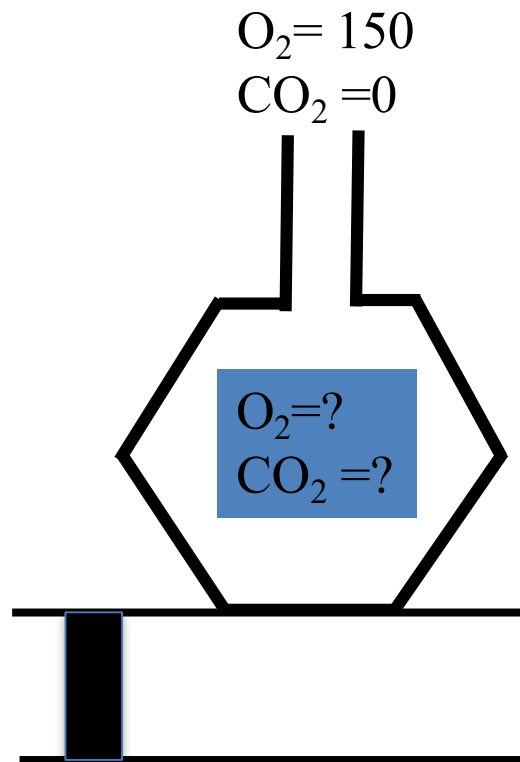
$O_2 = 159$
 $CO_2 = 0$



$$V/Q = 0.84$$

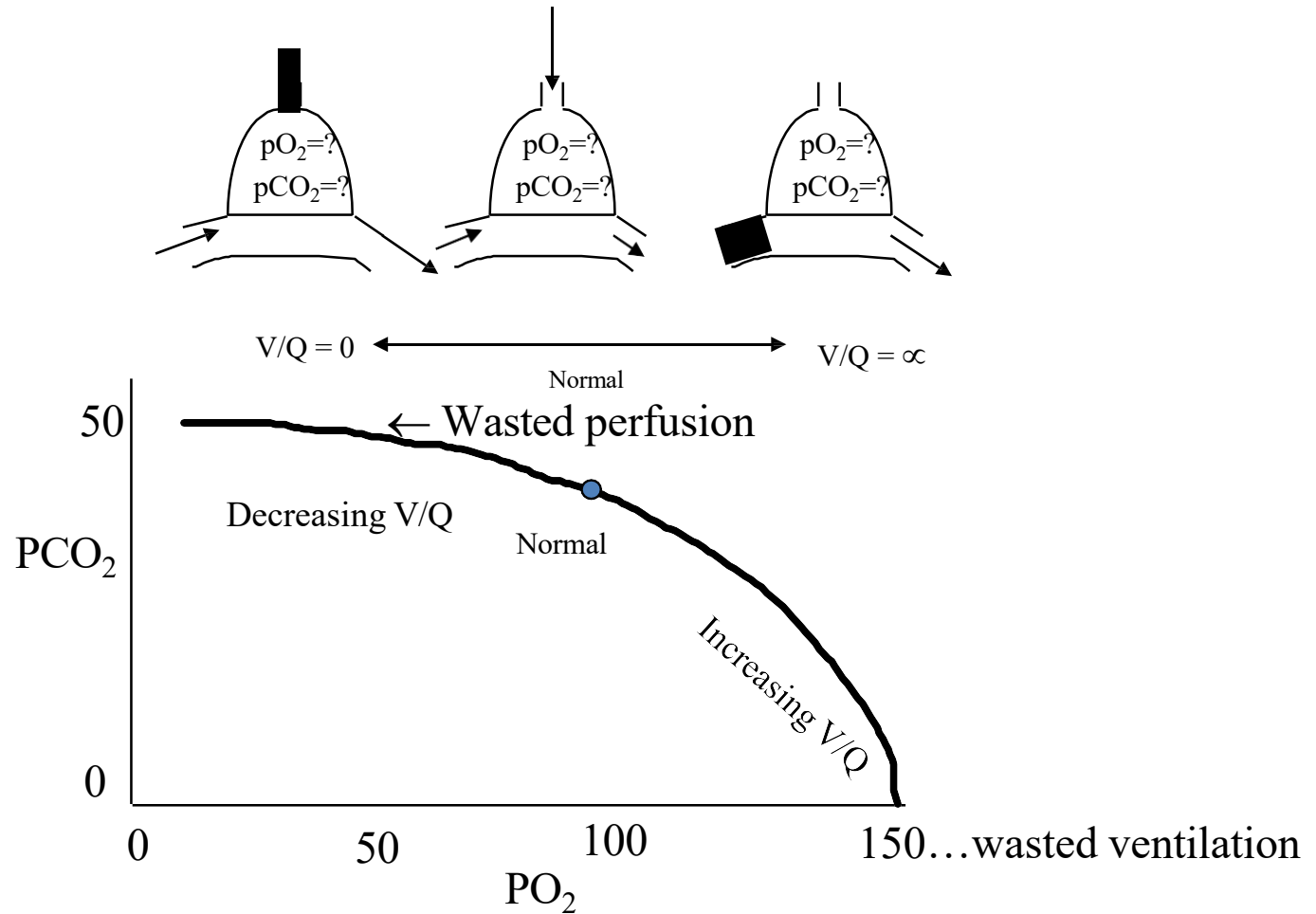


$$V/Q = 0$$



$$V/Q = \infty$$

Ventilation/perfusion



Abnormal VA/Q in the Upper and Lower Normal Lung.

- **Upper part of the lung**
 - Less blood flow and less ventilation; but blood flow is considerably less than ventilation.
 - Therefore, V/Q is 3.4 times higher than the normal value
 - This causes a moderate degree of *physiologic dead space*.
- **The bottom of the lung**
 - Slightly too little ventilation in relation to blood flow
 - V/Q as low as 0.6 times the normal value.
 - A small fraction of the blood fails to become normally oxygenated, and this represents a *physiologic shunt*.
 - Assuming perfusion is adequate ... hyperventilation makes alveolar air like atmospheric air Hypoventilation makes alveolar air like venous blood.

V/Q ratio

- Physiologic shunt
 - V/Q ratio is less than normal bcs of low ventilation
- Physiologic dead space...alveolar wasted volume
 - $V/Q > \text{normal}$

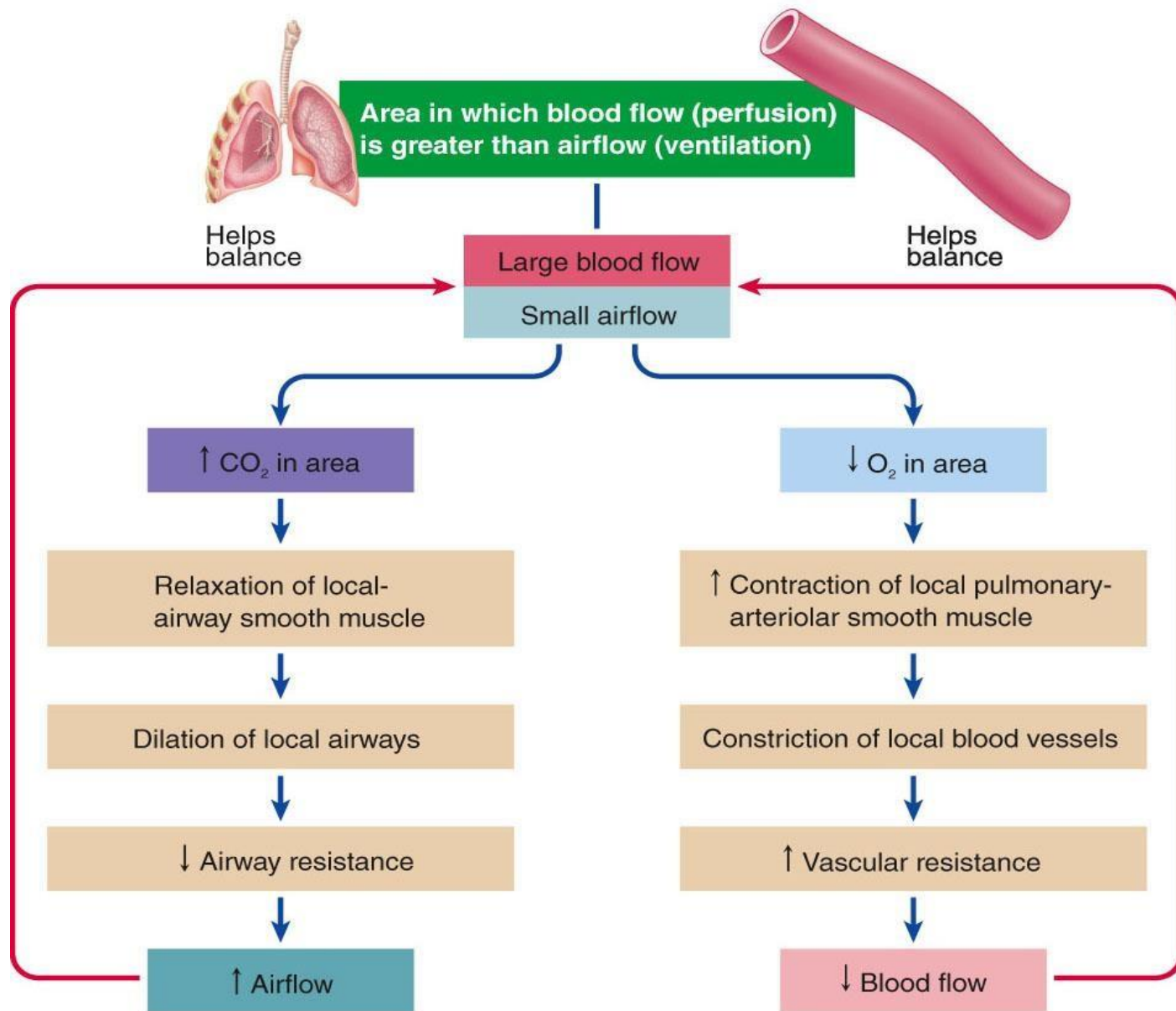
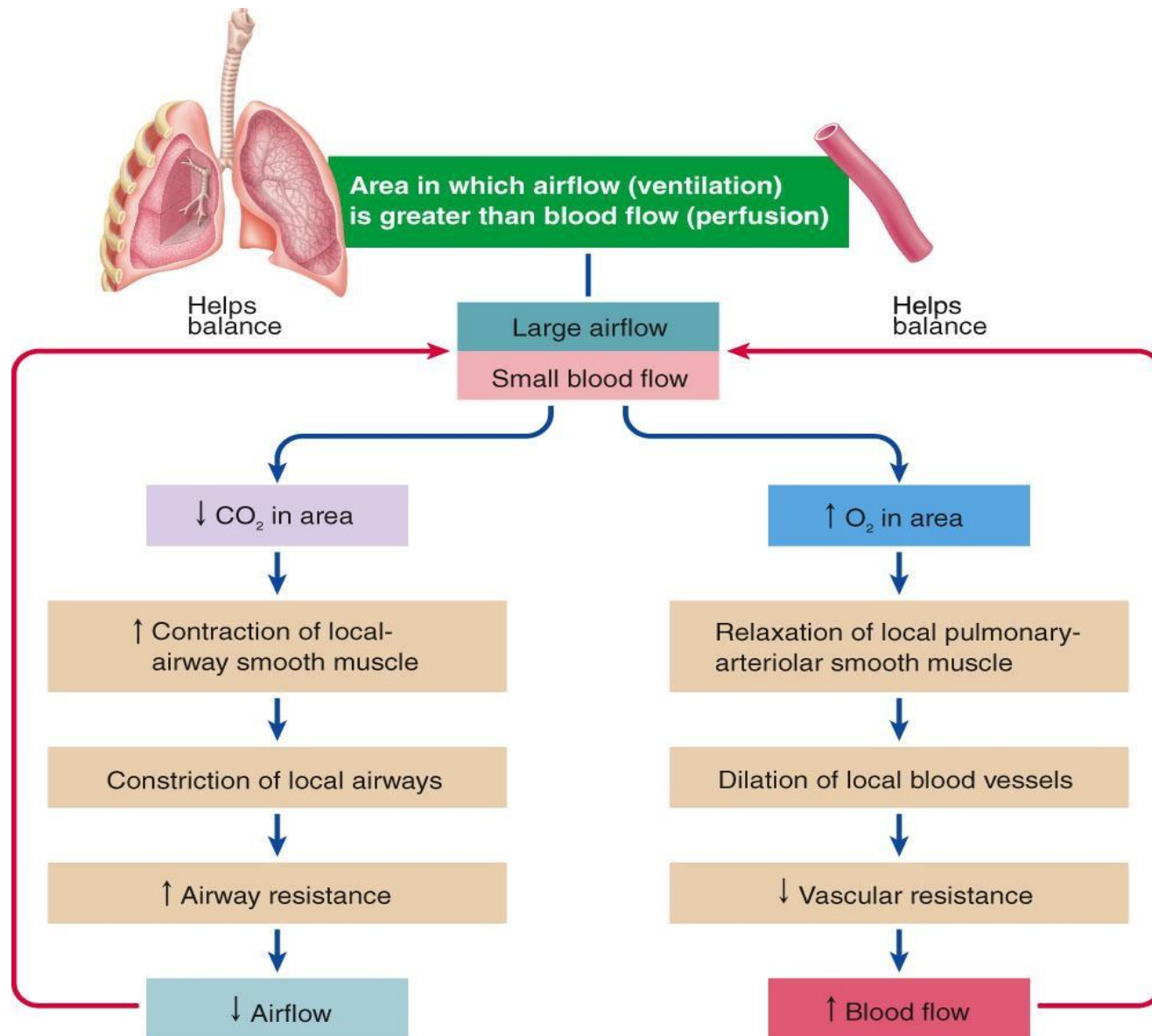


Fig Local controls to adjust ventilation and perfusion to lung area with large blood flow and small airflow



Fig

Local controls to adjust ventilation and perfusion to a lung area with large airflow and small blood flow

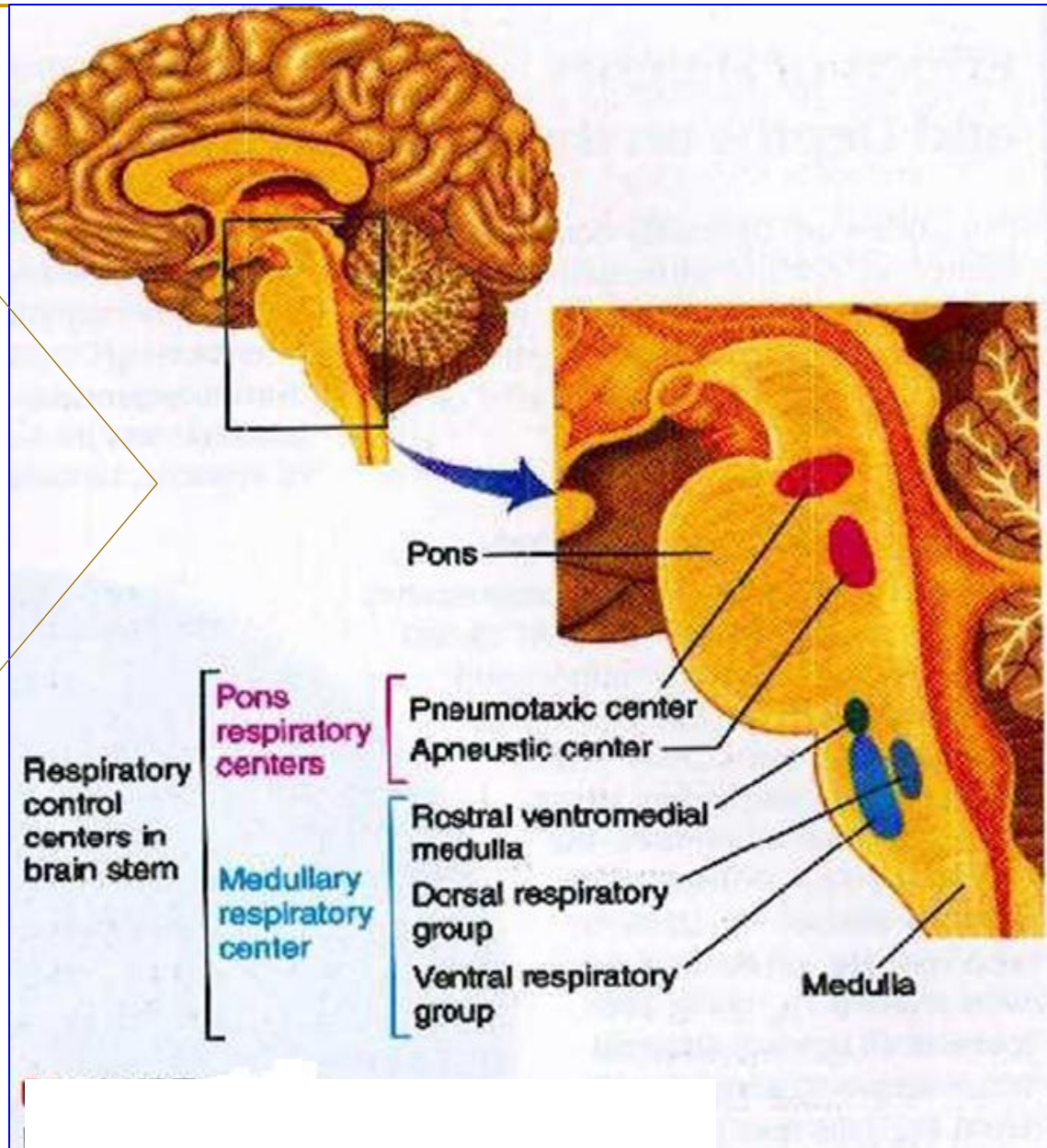
✓ Session 4:

Regulation of respiration

Regulation of respiration

- The two major respiratory centers in the **brain stem** are:
 - Medulla & Pons.
- Although the **basic rhythm of respiration** is set & coordinated by the **inspiratory area**, the rhythm can be modified in response to:
 - ❖ Inputs from other brain regions
 - ❖ Receptors in the peripheral nervous system &
 - ❖ Other factors

Respiratory centers in the brain stem



Regulation of respiration cont'

1) Medullary respiratory center

- Consists of dorsal respiratory group (DRG) and

Ventral respiratory group (VRG)

➤ DRG :

- ✓ Contain mostly inspiratory neurons (I-neurons).
- ✓ Responsible for rhythmic breathing.

➤ VRG :

- ✓ Contain expiratory neurons (E-neurons)

Regulation of Respiration cont'

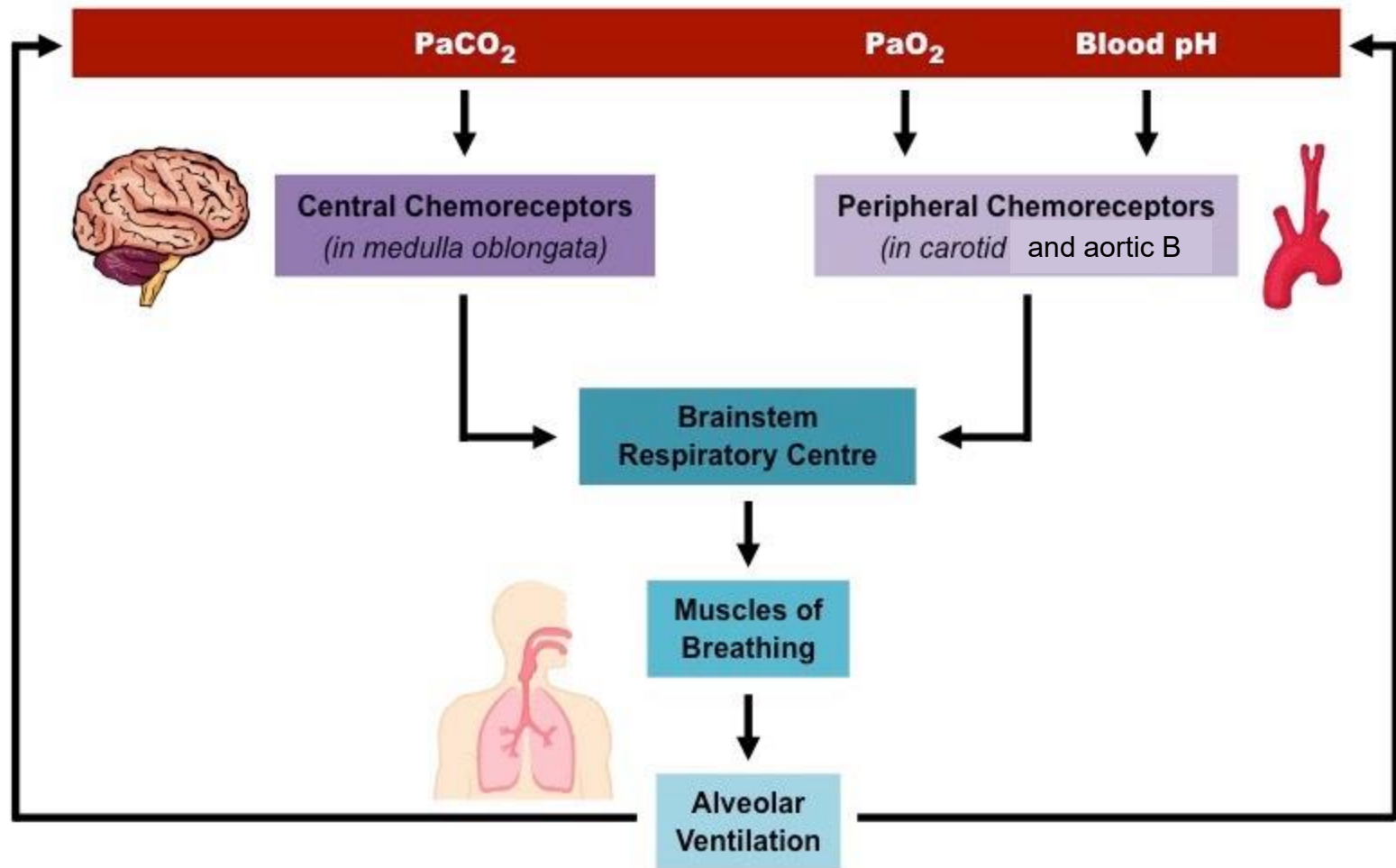
2) Pons respiratory center

- Influences output from medullary respiratory center.
 - **Pneumotaxic neurons**
 - ✓ **Switch off** I-neurons (limits duration of inspiration)
 - ✓ Shorten the duration of inhalations & ↑ breathing rate
 - **Apneustic neurons**
 - ✓ **Prevent** I-neurons from being **switched off**
 - ✓ Activate inspiratory area & prolong inhalation, result in long, deep inhalation
- Pneumotaxic dominant over apneustic, allowing for smooth breathing.

Regulation of respiration cont'

- Control of ventilation by PO_2 , PCO_2 & H^+ .
- Achieved via **2 chemoreceptors** :
 1. Peripheral chemoreceptors.
 2. Central chemoreceptors.
- Controls breathing via nerve fibers to the respiratory control centers.

Regulation of respiration cont'

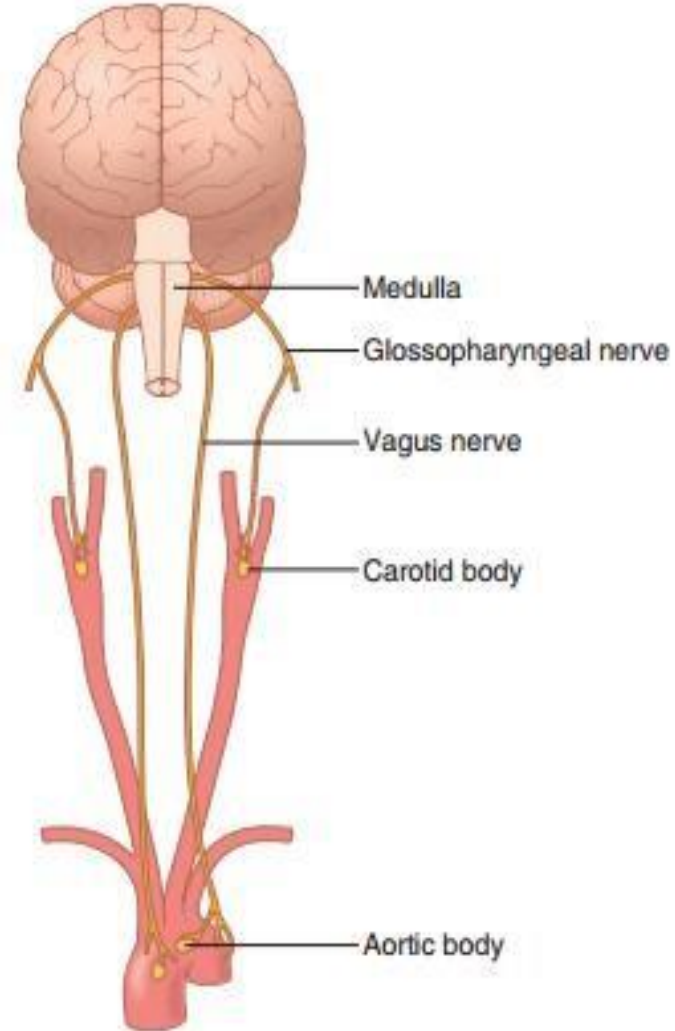


Peripheral chemoreceptors

- located in the **carotid bodies** & **aortic bodies**.

Central chemoreceptors

- located on the ventral surface of the **medulla**.



Regulation of Respiration cont'

1. Peripheral chemoreceptors

- **Sense changes** in arterial O_2 , CO_2 & H^+ .
- $\uparrow PCO_2 \rightarrow$ chemoreceptor $\rightarrow$ generation of AP $\rightarrow$ conduction of AP via sensory neurons $\rightarrow$ respiratory **control center** $\rightarrow$ motor out put via motor neurons $\rightarrow$ effectors (respiratory muscle) $\rightarrow$ $\uparrow$ ventilation (CO_2 blown off) $\rightarrow$ $\downarrow PCO_2$.
- $\uparrow H^+ \rightarrow$ chemoreceptor $\rightarrow$ respiratory control center $\rightarrow$ $\uparrow$ ventilation $\rightarrow$ $\downarrow PCO_2 \rightarrow \downarrow H^+$.

Regulation of Respiration cont'

- Control of respiration in **mammals** is regulated by changes in **PCO₂** (not **PO₂**).
 - Peripheral O₂ chemoreceptors do not contribute in regulating normal ventilation unless **arterial PO₂** falls below **60 mm Hg**.
- Peripheral O₂, CO₂ & H⁺ chemoreceptors are weakly responsive & play a minor role in controlling respiration.

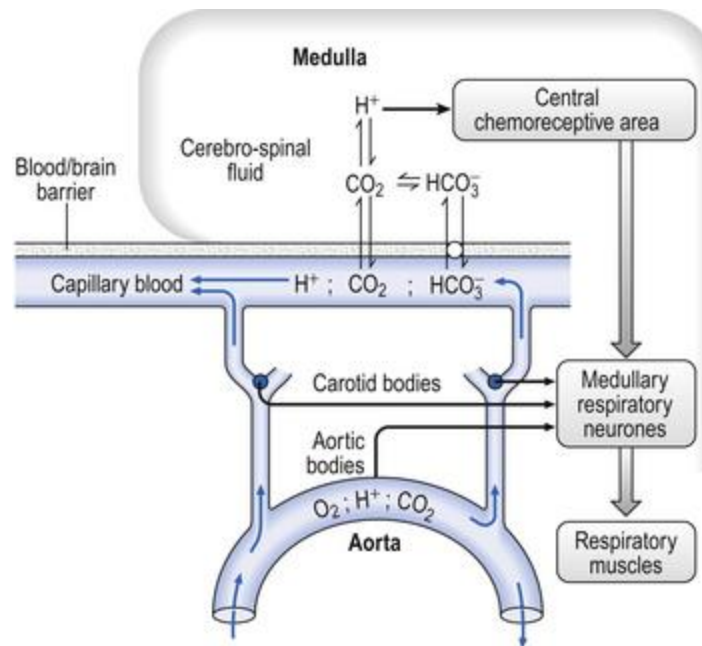
Regulation of Respiration cont'

2. Central chemoreceptors

- Most important regulator of ventilation.
- **Do not monitor** changes in **PCO₂ directly**.
- Respond to changes in **CO₂-induced** production of **H⁺** in cerebrospinal fluid (brain interstitial fluid).
- **Blood-brain barrier** allows the diffusion of CO₂ but is **impermeable to H⁺**.

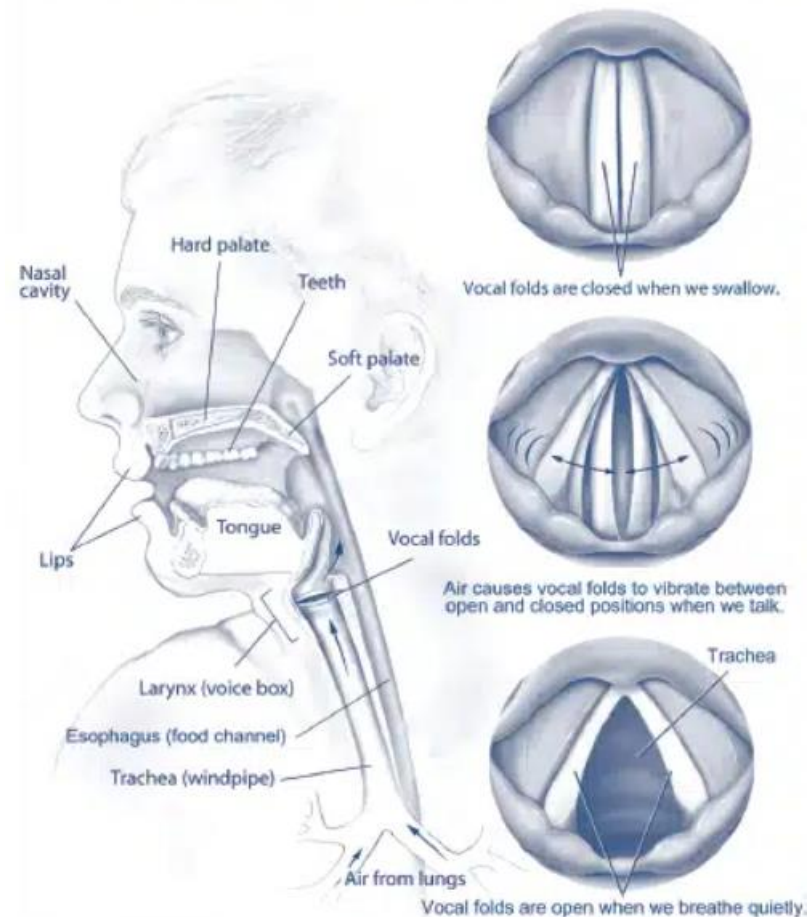
Regulation of Respiration cont'

2. Central chemoreceptors

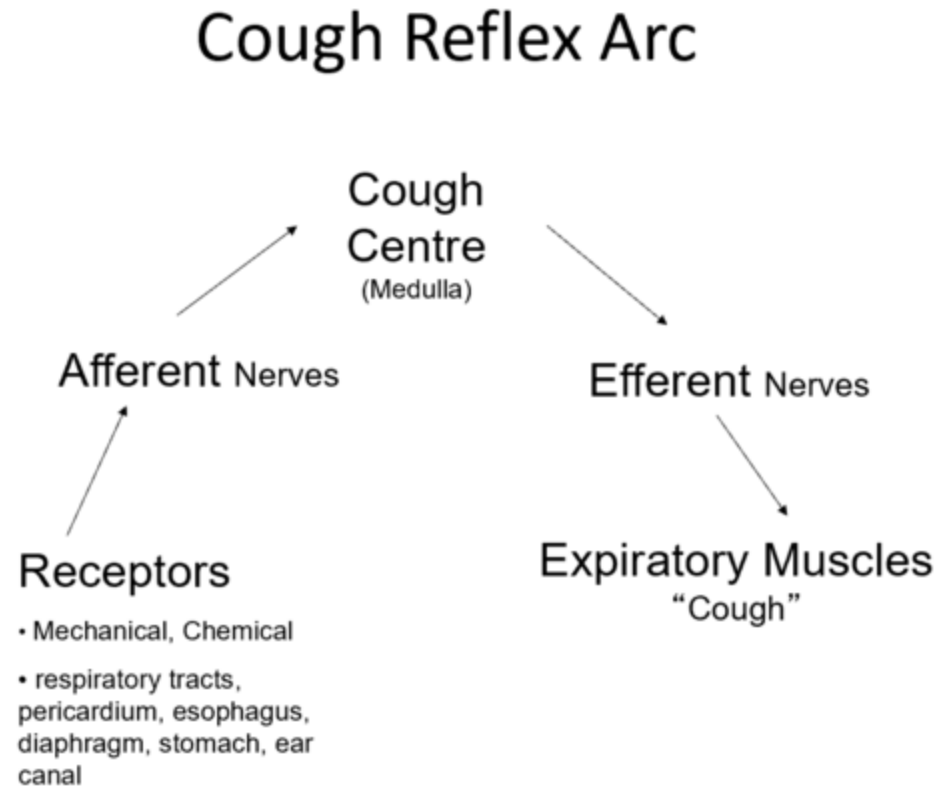


Cough Reflex

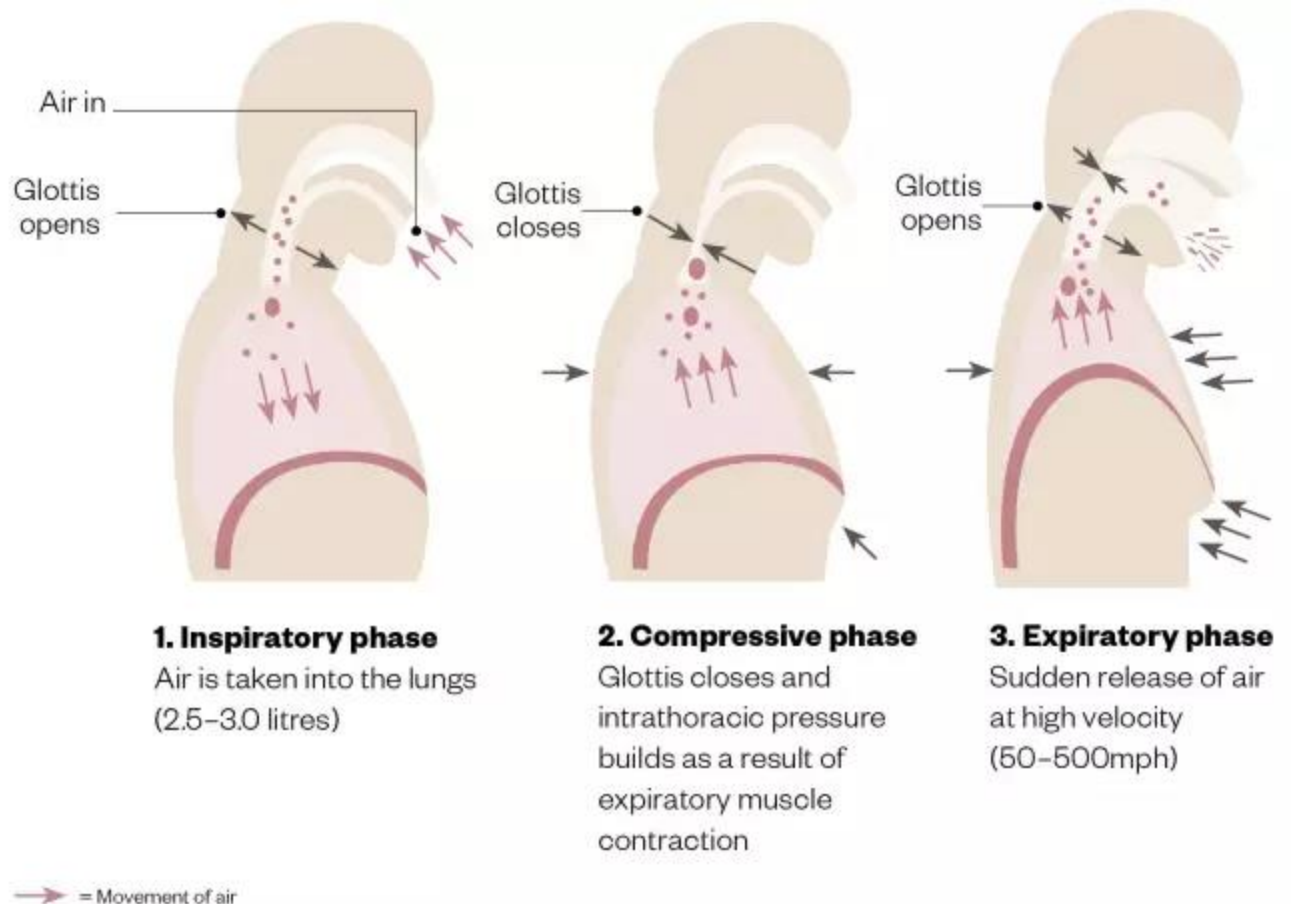
- The cough reflex is an important defence mechanism which clears the airways of irritants by forcefully expelling air from the **respiratory tract**.
- Phases of cough reflex



Cough Reflex Arc

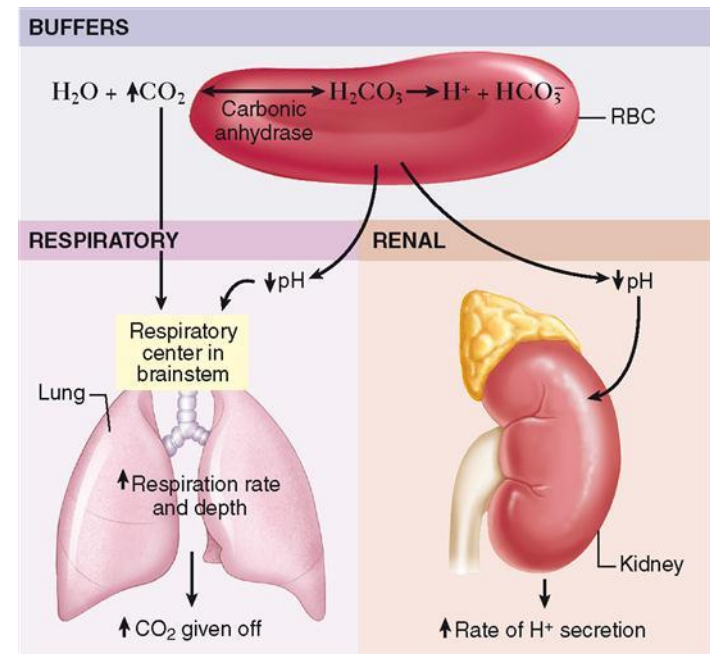


Phase of Cough Reflex



Respiratory Regulation of Acid-Base Balance

- The acid-base balance is vital for normal bodily functions.
- Disruption of this equilibrium can lead to severe complications.
- Therefore, this balance is tightly regulated.
- In addition to renal and chemical buffer system.
- Respiratory system is important for Acid base balance.



Thank you!
